

Introduction

CS524: Big Data Visualization & Analytics
Fabio Miranda



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evl

electronic
visualization
laboratory



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Big Data Visualization & Analytics



Big Data Visualization & Analytics

What is big data?

What is big data?

Volume:
Enormous
quantities of data

Velocity:
Created in real-
time

Variety
Structured, semi-
structured, and
unstructured

[Doug Laney, 2001]

What is big data?

Volume:

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Velocity:

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Variety:

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Exhaustivity:

An entire system is captured rather than being sampled
(Mayer-Schonberger and Cukier, 2013)

Fine-grained in resolution and **uniquely indexical** in identification
(Dodge and Kitchin, 2005)

Relationality:

Contains common fields that enable the conjoining of different datasets
(Boyd and Crawford, 2012)

Extensionality (can add new fields easily) and **scalability** (can expand in size rapidly)
(Marz and Warren, 2012)

Veracity:

The data can be messy, noisy, and contain uncertainty and error
(Marr, 2014)

Value:

Many insights can be extracted and the data repurposed
(Marr, 2014)

Variability:

Data whose meaning can be constantly shifting in relation to the context in which they are generated
(McNulty, 2014)

[Kitchin and McArdle, 2016]



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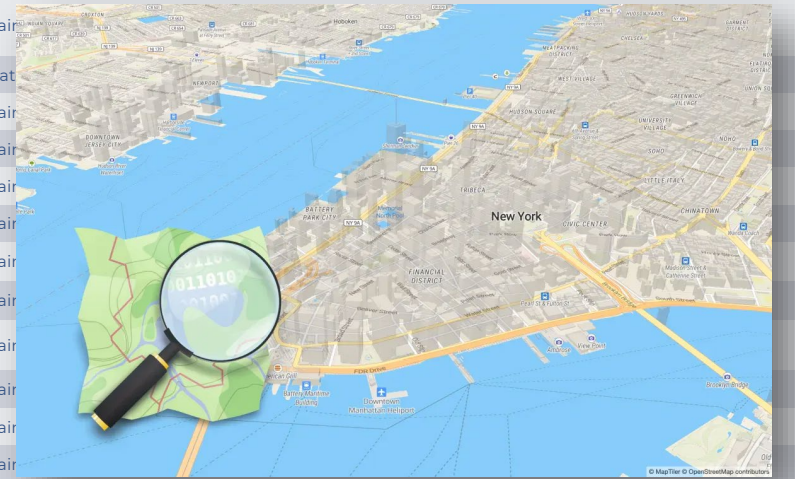


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Category	Data Source	Volume (Records)	Volume per Record	Volume (TBs/PBs)	Velocity (Generation)	Velocity (Handling)	Variety	Exhaustivity	Resolution	Indexical	Relational	Extensionality	Scalable
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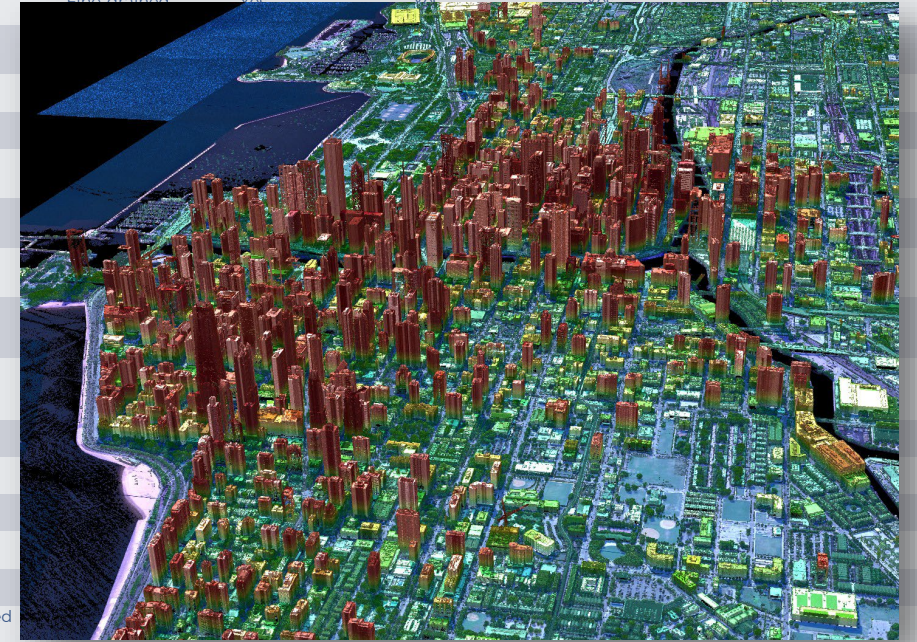
[Kitchen and McArdle, 2016]

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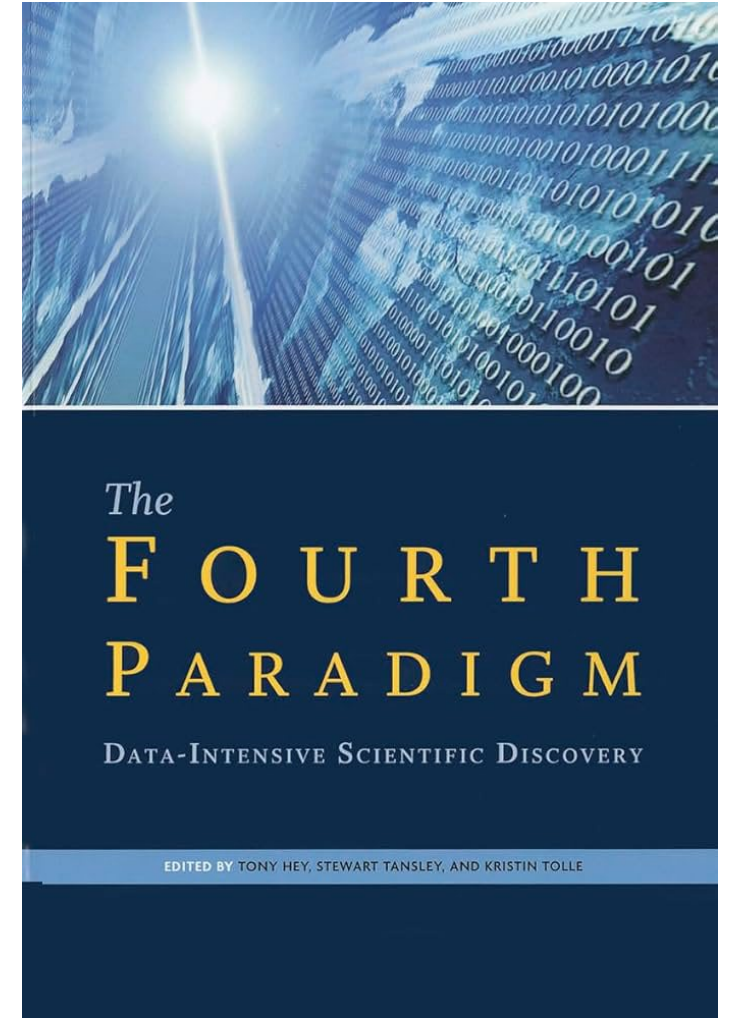
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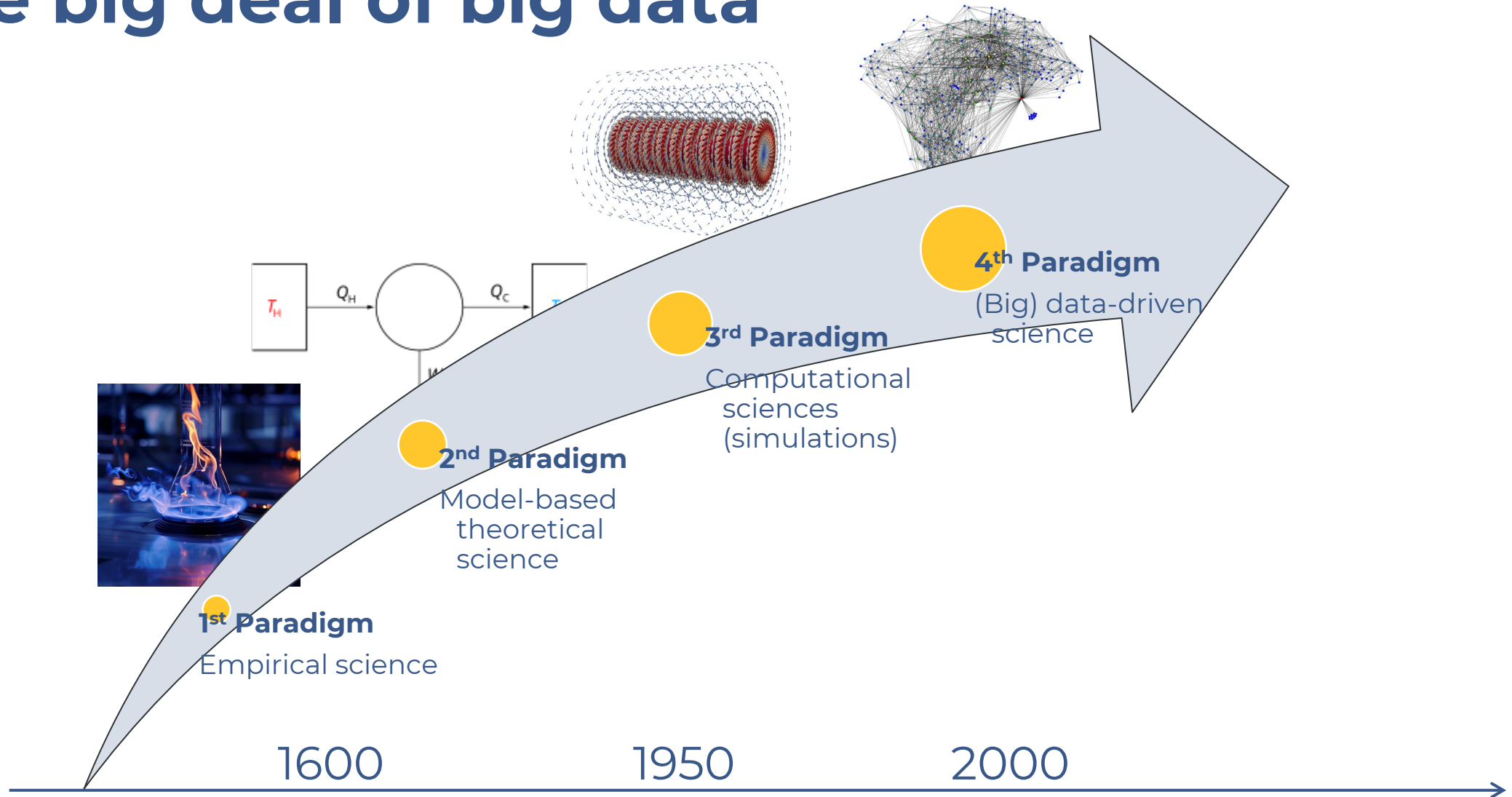
[Kitchen and McArdle, 2016]

The big deal of big data

- **Science**: several domains are moving toward data-driven exploration: *increasing use of data is bringing a paradigm shift to the nature of science.*
 - Climate data, urban data, physics data, health data.
- **Industry**: companies are capitalizing on data – users are consuming and producing data.
 - Social media data, crowdsourced data, sensor data.
- **Government**: agencies use data to operate efficiently, make policies, make informed decisions.
 - Data.gov: 224,669+ datasets; NYC Open Data: 1,400 datasets.



The big deal of big data



Big Data **Visualization & Analytics**

What is visualization & visual analytics?

What is visualization & visual analytics?

- Visualization & visual analytics are not just “making charts.”
- Visualization: Graphical representation of information and data.
- Visual analytics: Science of analytical reasoning facilitated by interactive visual interfaces, integrating automated analysis (AI / ML) with human expertise.

Google Scholar

Top publications

Categories > Engineering & Computer Science > Computer Graphics

Publication	h5-index	h5-median
1. ACM Transactions on Graphics (TOG)	114	177
2. IEEE Transactions on Visualization and Computer Graphics	94	132
3. Computer Graphics Forum	64	95
4. The Visual Computer	63	87
5. Computers & Graphics	38	59
6. SIGGRAPH Asia	37	76
7. IEEE Computer Graphics and Applications	27	42
8. ACM Symposium on Virtual Reality Software and Technology	27	37
9. Visual Informatics	25	40
10. Proceedings of the ACM on Computer Graphics and Interactive Techniques	23	39
11. IEEE Visualization Conference	23	32
12. Computer Aided Geometric Design	23	28
13. SIGGRAPH Conference on Graphics, Patterns and Images	22	37
14. IEEE Pacific Visualization Symposium	19	36
15. Computer Animation and Virtual Worlds	19	26
16. Graphical Models	18	29
17. International Journal of Image and Graphics	18	24
18. International Conference on Information Visualization	15	21
19. Symposium on Graph Drawing (GD)	13	19
20. Graphics Interface Conference	13	18

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VIS2026

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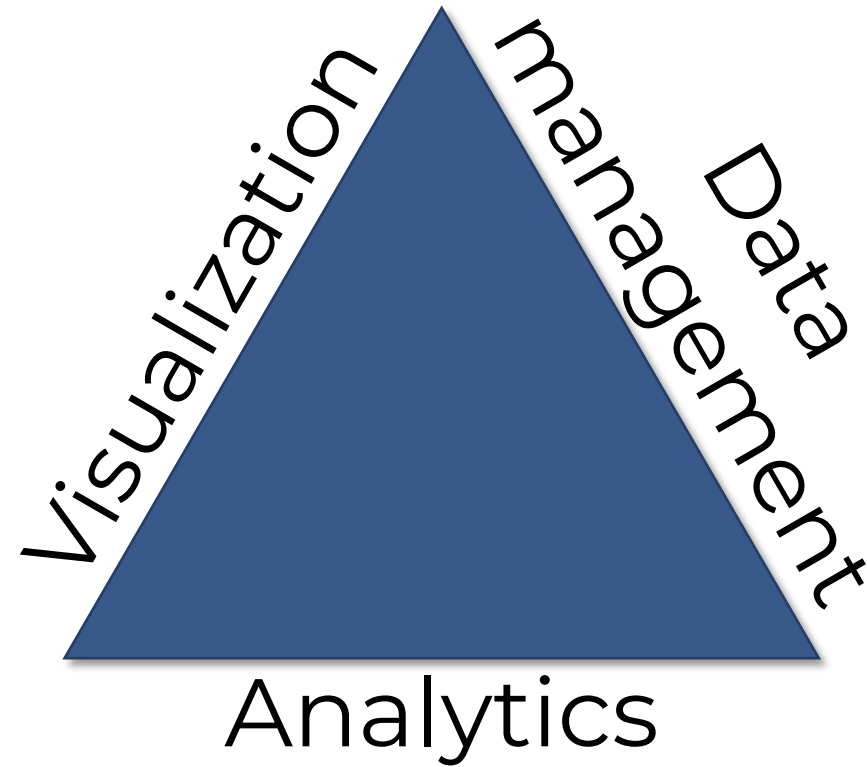
IEEE VIS 2026 in Boston, Massachusetts



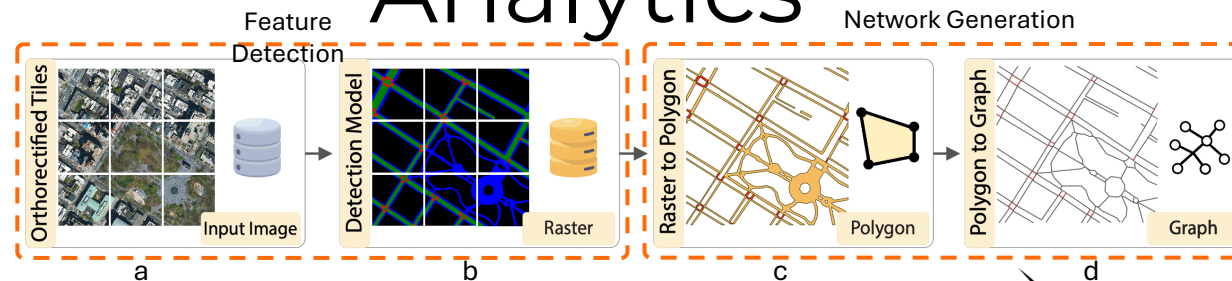
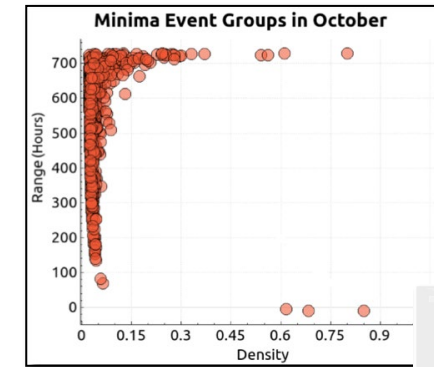
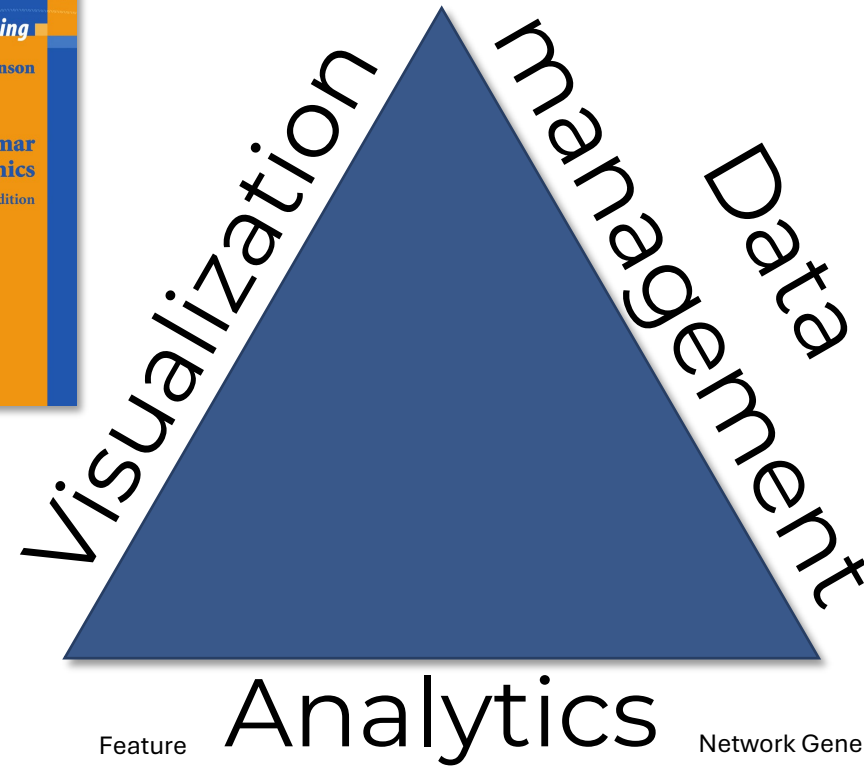
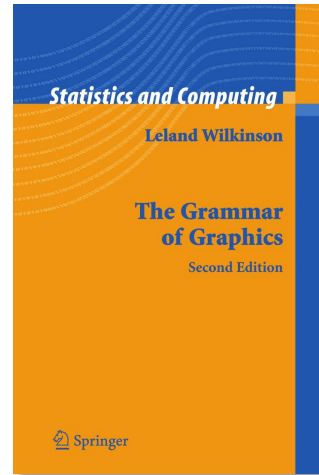
What is visualization & visual analytics?



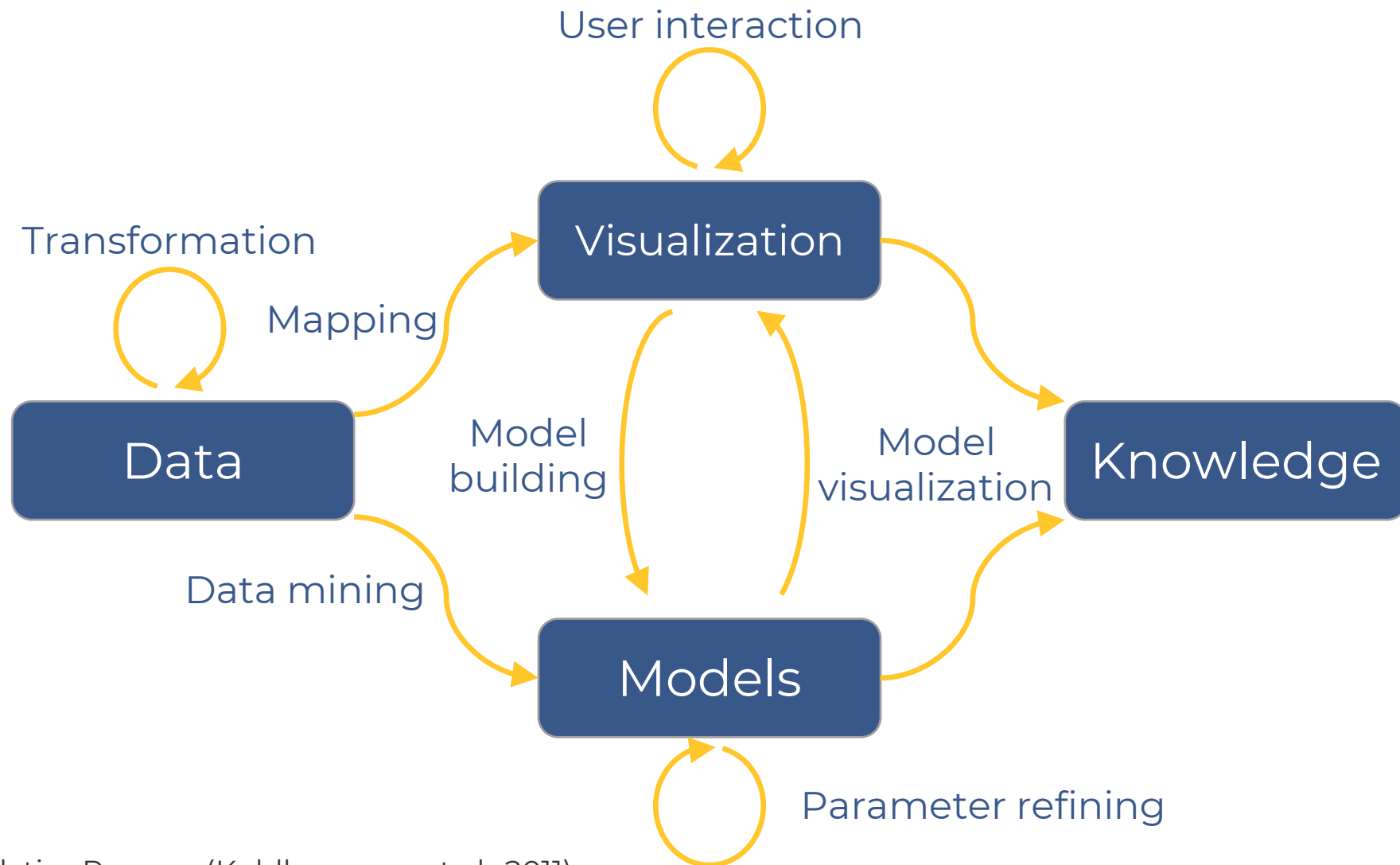
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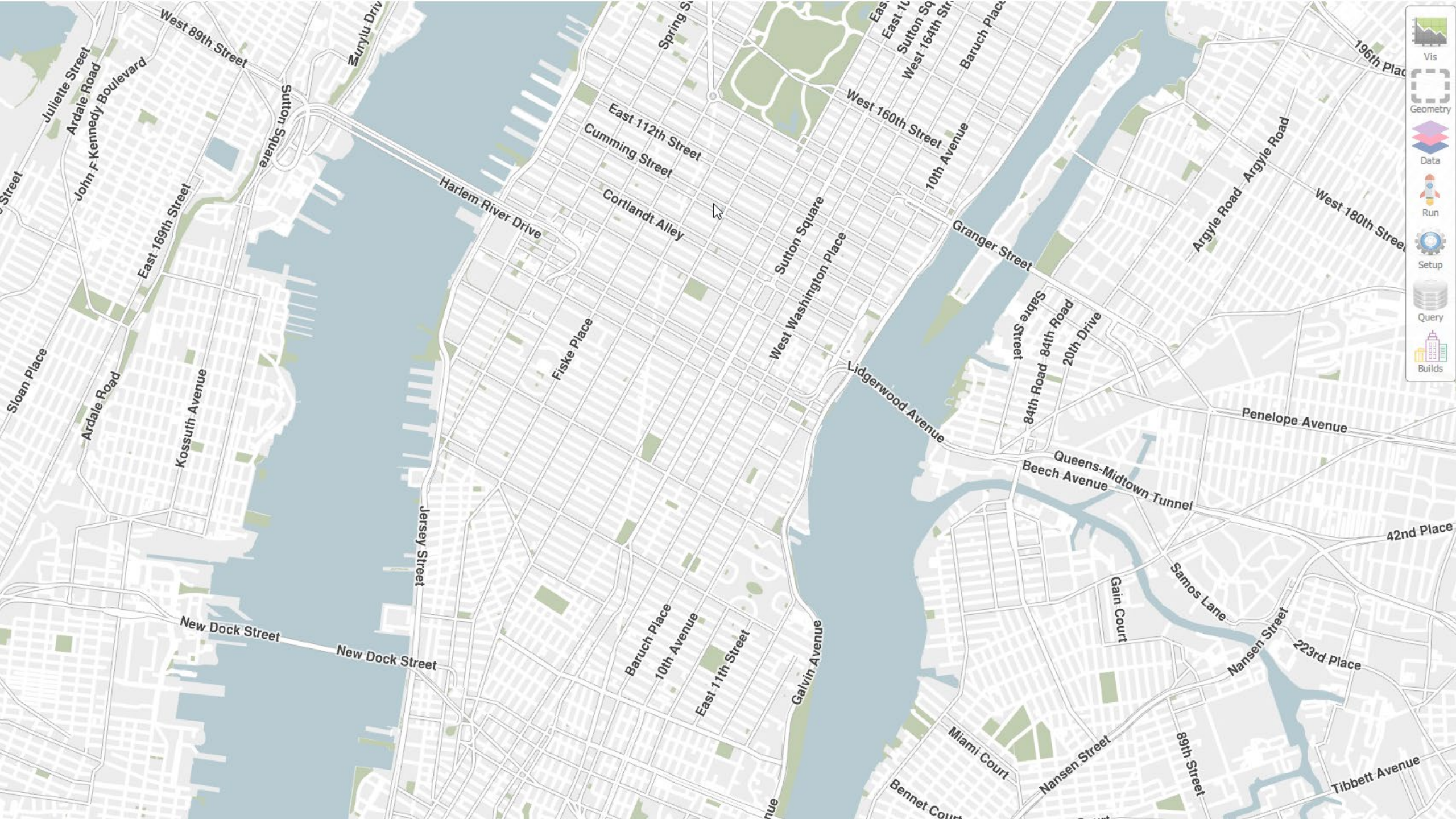
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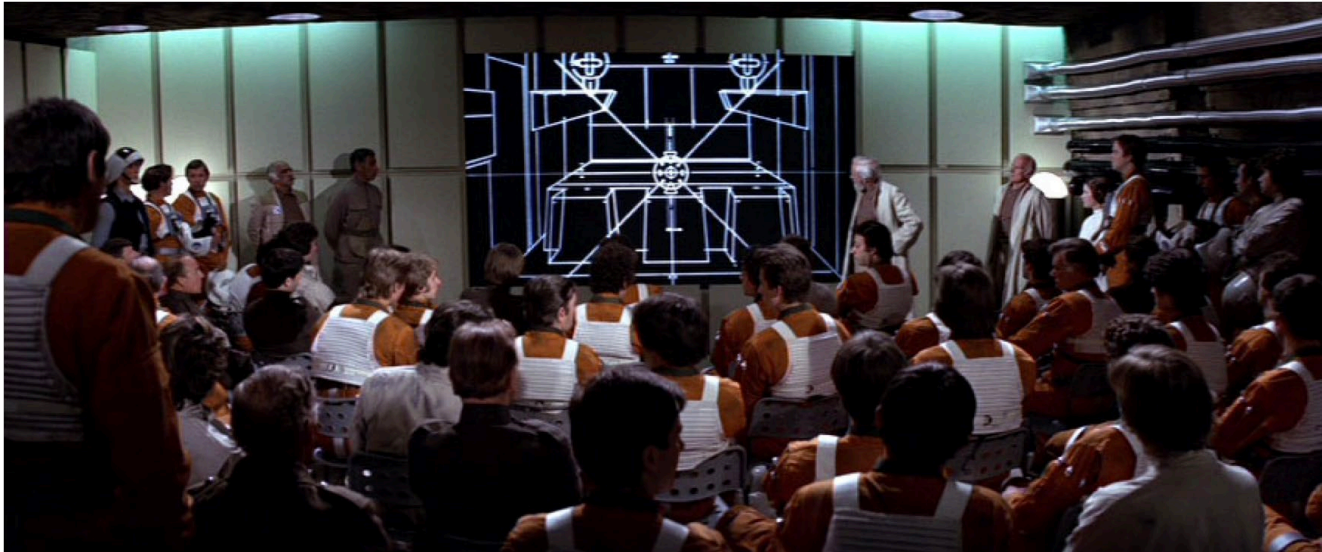
The Visual Analytics Process (Kohlhammer et al., 2011)



-  Vis
-  Geometry
-  Data
-  Run
-  Setup
-  Query
-  Builds

Visualization at UIC: EVL

- EVL: established in 1973 by Tom DeFanti & Dan Sandin.
- Fundamental in the creation of the animated Death Star sequence for Star Wars.



Visualization at UIC: EVL

Maxine D. Brown from EVL edited the 1987 NSF-sponsored report “Visualization in Scientific Computing” outlining the emerging field of scivis.

Visualization in Scientific Computing

Editors

Bruce H. McCormick
Texas A&M University
College Station, TX

Thomas A. DeFanti
University of Illinois at Chicago
Chicago, IL

Maxine D. Brown
University of Illinois at Chicago
Chicago, IL

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Any opinions, findings and conclusions or recommendations expressed in this publication are those of the authors and do not necessarily reflect the views of the National Science Foundation.

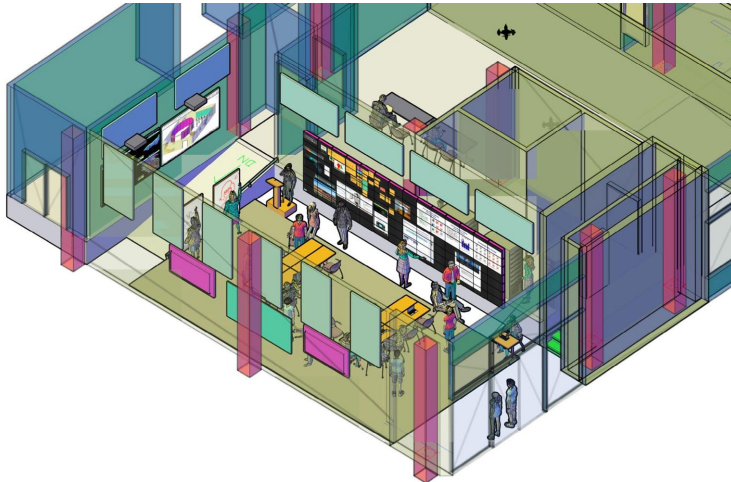
Published July, 1987.

Visualization at UIC: EVL



Visualization at UIC: EVL

- DOCC: Visual Data Science space
- Successor to EVL approach:
 - CAVE2 - VR Hybrid Space
 - Continuum - Instrumented Room
 - COMPaaS – Adaptive GPU Cluster
- VR
- Interactive AI
- Collaboration Space



Large Stereo 3D wall
2x GPU SuperServers for
Foundation AI models
Jupyter with GPUs on demand
100Gbps network
Integrated video conferencing
3D Tracking for VR
Ancillary displays and tech



Visualization at UIC: EVL

The Department of Computer Science is ranked #16 in csrankings.org in the area of **Visualization**.

CSRankings: Computer Science Rankings

CSRankings is a metrics-based ranking of top computer science institutions around the world. [Click on a triangle \(▶\)](#) to expand areas or institutions. [Click on a name](#) to go to a faculty member's home page. [Click on a chart icon](#) (the ) after a name or institution) to see the distribution of their publication areas as a [bar chart](#). [Click on a Google Scholar icon](#) (🔍) to see publications, and [click on the DBLP logo](#) (DBLP) to go to a DBLP entry. [Applying to grad school? Read this first.](#) For info on grad stipends, check out [CSStipendRankings.org](#). For publication statistics on CSRankings conferences, see [CSconferences.org](#). If you find CSRankings useful, [consider sponsoring CSRankings](#).

Rank institutions in USA by publications from 2015 to 2025

All Areas [\[off | on\]](#)

AI [\[off | on\]](#)

- ▶ Artificial intelligence ☐
- ▶ Computer vision ☐
- ▶ Machine learning ☐
- ▶ Natural language processing ☐
- ▶ The Web & information retrieval ☐

Systems [\[off | on\]](#)

- ▶ Computer architecture ☐
- ▶ Computer networks ☐
- ▶ Computer security ☐
- ▶ Databases ☐
- ▶ Design automation ☐
- ▶ Embedded & real-time systems ☐
- ▶ High-performance computing ☐
- ▶ Mobile computing ☐
- ▶ Measurement & perf. analysis ☐
- ▶ Operating systems ☐
- ▶ Programming languages ☐
- ▶ Software engineering ☐

Theory [\[off | on\]](#)

- ▶ Algorithms & complexity ☐
- ▶ Cryptography ☐
- ▶ Logic & verification ☐

Interdisciplinary Areas [\[off | on\]](#)

- ▶ Comp. bio & bioinformatics ☐
- ▶ Computer graphics ☐
- ▶ Computer science education ☐
- ▶ Economics & computation ☐
- ▶ Human-computer interaction ☐
- ▶ Robotics ☐
- ▼ Visualization ☒

IEEE VGTC

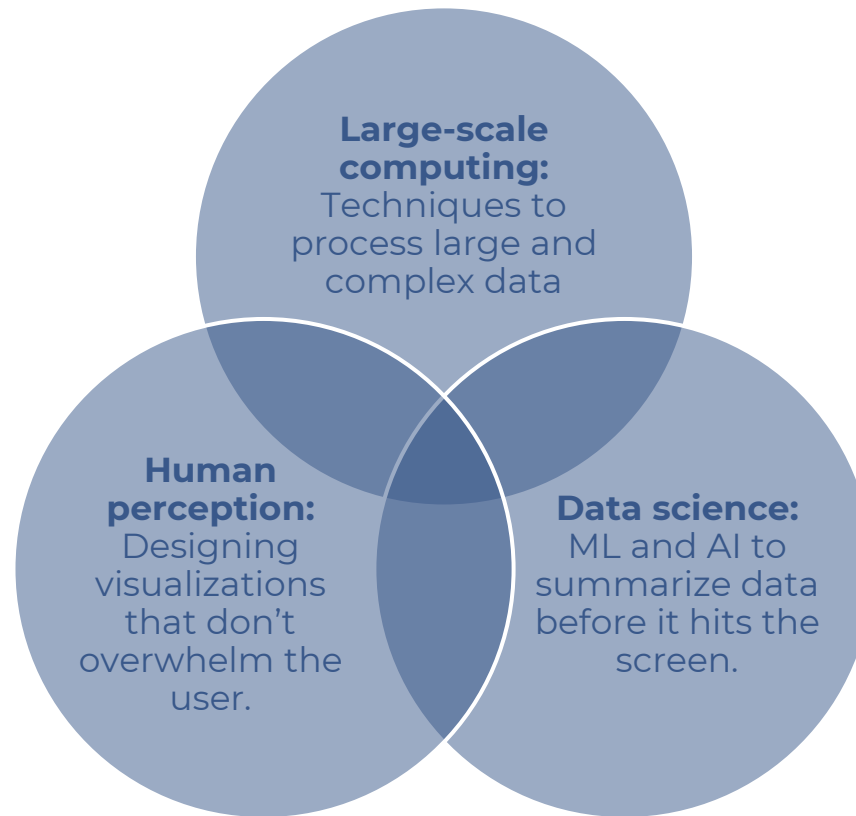
- VIS ☒
- VR ☐

Institution

- ▶ University of Utah 
- ▶ Northwestern University 
- ▶ Georgia Institute of Technology 
- ▶ Northeastern University 
- ▶ Univ. of California - Davis 
- ▶ Ohio State University 
- ▶ Harvard University 
- ▶ University of Washington 
- ▶ University of North Carolina 
- ▶ Oregon State University 
- ▶ University of Minnesota 
- ▶ Stony Brook University 
- ▶ New York University 
- ▶ Arizona State University 
- ▶ University of Maryland - College Park 
- ▶ University of Illinois at Chicago 
- ▶ University of Michigan 
- ▶ University of Chicago 
- ▶ University of Oklahoma 
- ▶ University of Wisconsin - Madison 
- ▶ Virginia Tech 
- ▶ Tufts University 
- ▶ University of Arizona 

Now... what is big data visualization & analytics?

Big data vis & analytics is the practice of analyzing and visualizing datasets and models that are so massive, fast-moving, or complex that traditional database and visualization tools fail.



Now... what is big data visualization & analytics?

- Big data raises several challenges to visualization & visual analytics, e.g.:
 - **Volume**
 - Overplotting: If you plot 1 billion points, you just get a black square. You must aggregate or sample.
 - Memory & Rendering Bottlenecks: Standard hardware cannot store or render petabytes of data at 60 frames per second.
 - **Velocity**
 - Real-time Latency: If data updates every millisecond, the visualization must be "streaming." If the system lags by even a second, the analyst is making decisions based on the "past," not the "now."
 - **Variety**
 - Schema Heterogeneity: How do you visualize a single dashboard that contains unstructured text (tweets), structured tables (SQL), and binary images (X-rays) all at once?
 - High-Dimensionality: Big data often has hundreds of variables (columns). Our eyes can only see 2D or 3D, so we face the "Curse of Dimensionality" when trying to map complex data to a simple screen.
 - **Veracity**
 - Uncertainty Visualization: Big data is often noisy or missing pieces. The challenge is showing the probability of an error without making the map look messy.
 - Provenance/Lineage: If the AI intern cleans or filters the data, the human needs to see the "receipts." We must visualize the steps taken to ensure the data hasn't been biased by the algorithm.
 - **Value**
 - Low Signal-to-Noise Ratio: In the "Fifth Era," 99% of the data might be irrelevant noise. The challenge is designing "Smart Filters" that highlight the 1% that leads to a breakthrough.



Now... what is big data visualization & analytics?

- Scalable tools and techniques that help domain experts find, clean, integrate, ***interactively*** explore, and explain data.
- Guide users in the exploration process.
- Support interactive queries:
 - *“increased latency reduces the rate at which users make observations, draw generalizations, and generate hypotheses”.*

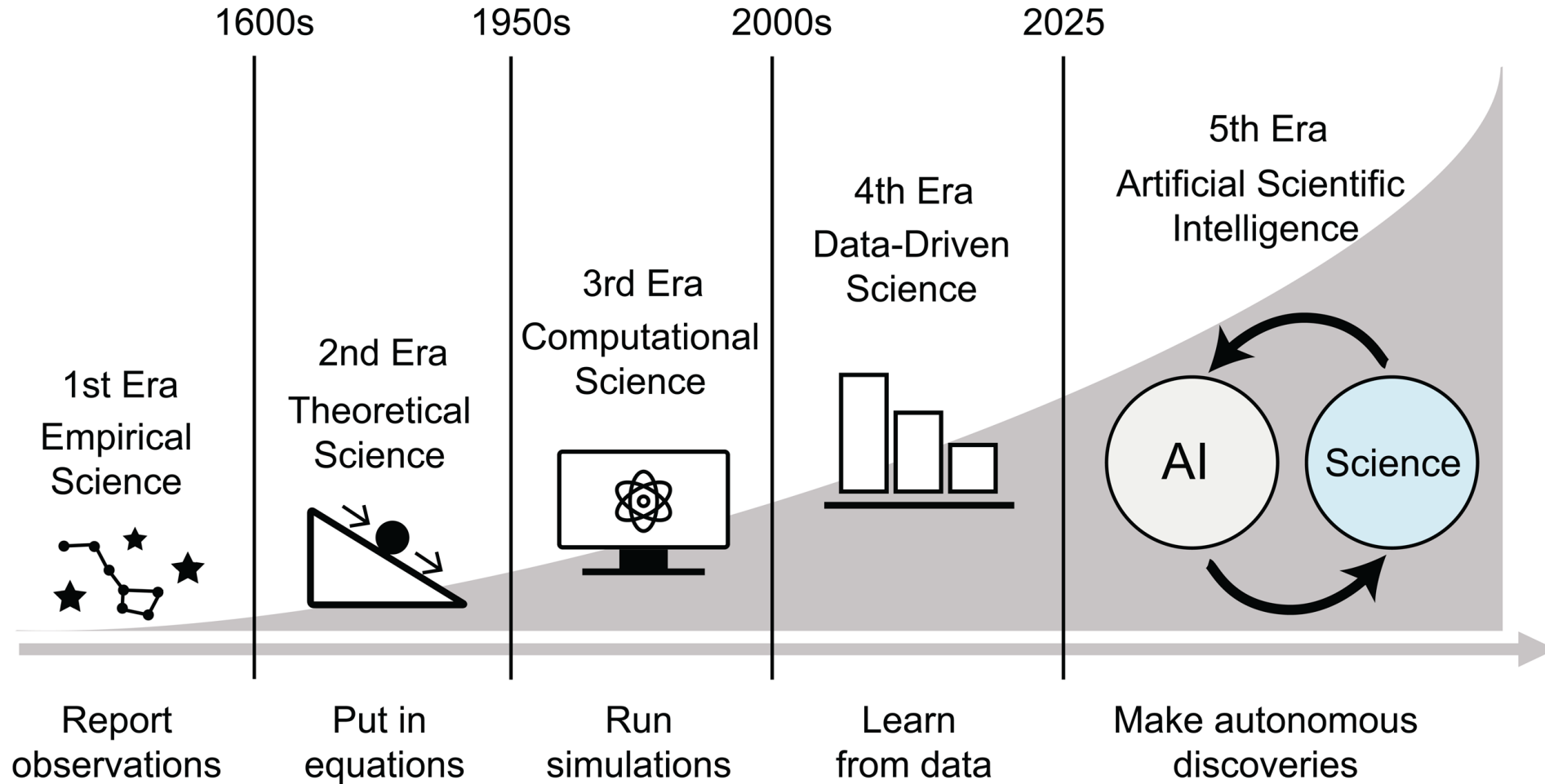
[Liu and Heer, IEEE TVCG 2014]

- Interdisciplinary:
 - *“as data scale and complexity increases, the novel solutions that will ultimately enable interactive, large-scale exploratory data analysis will have to come from truly interdisciplinary work”.*

[Chang, Fekete, Freire and Scheidegger, Dagstuhl Reports 2017]

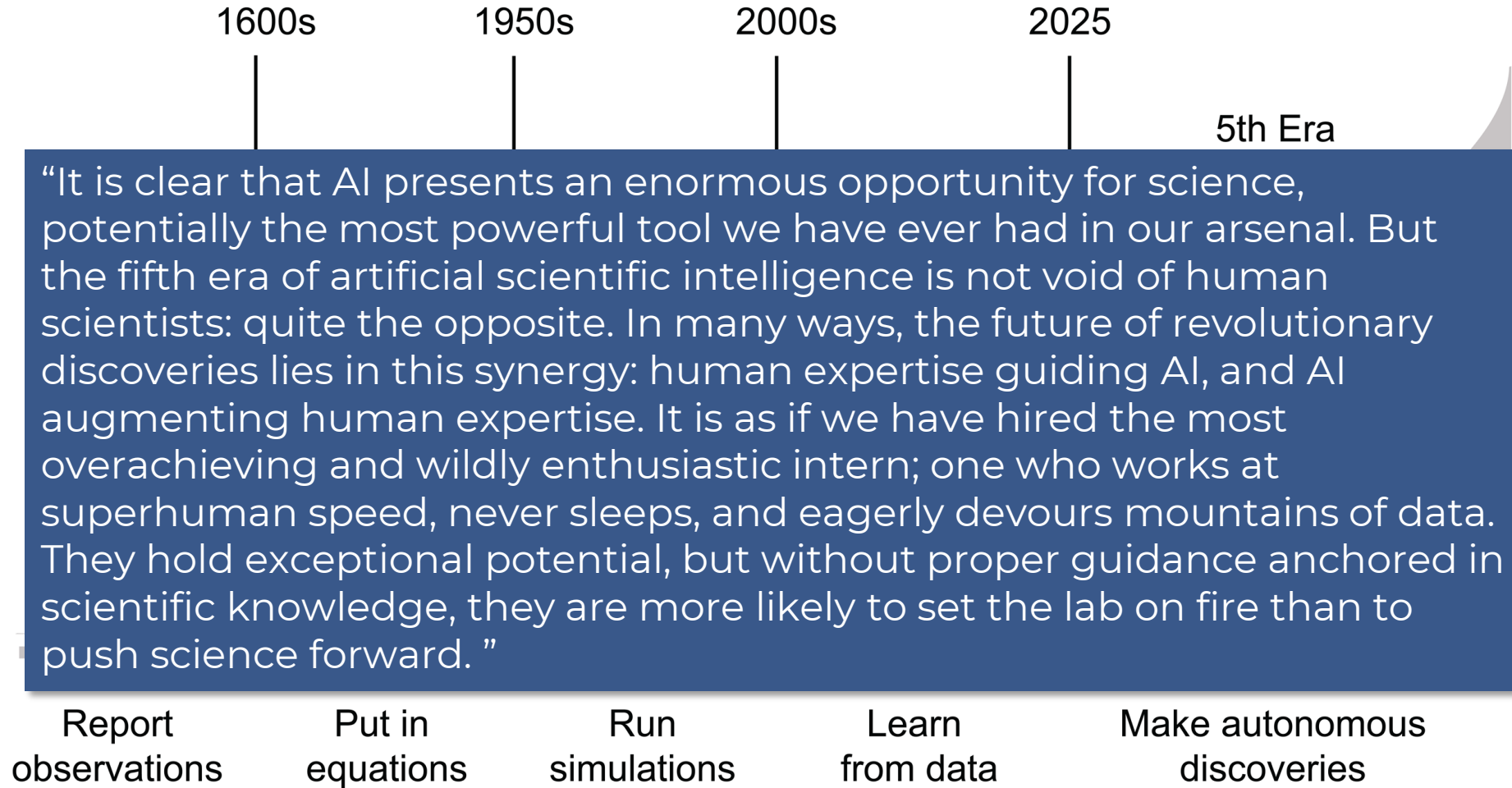


The big deal of big data



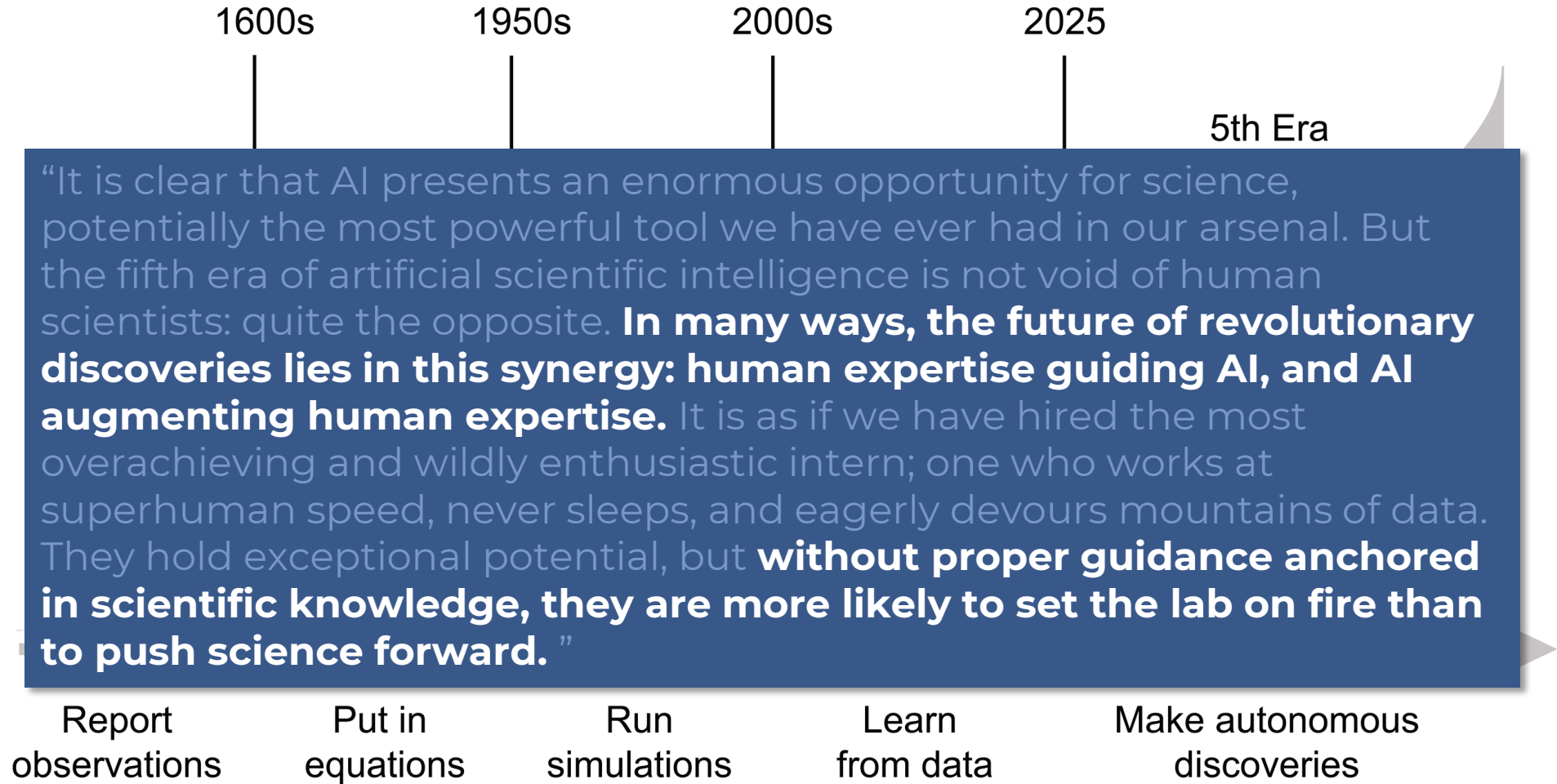
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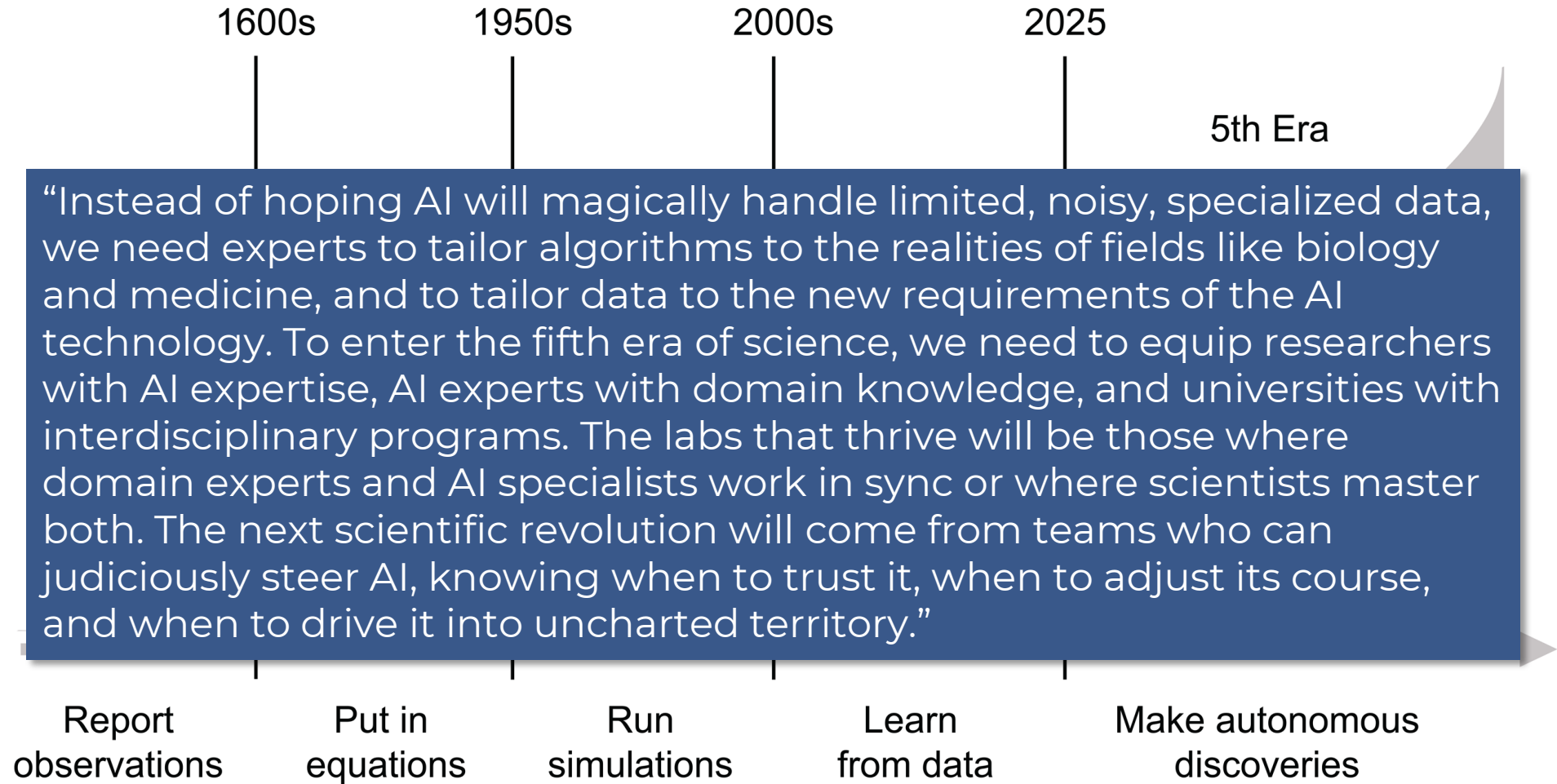
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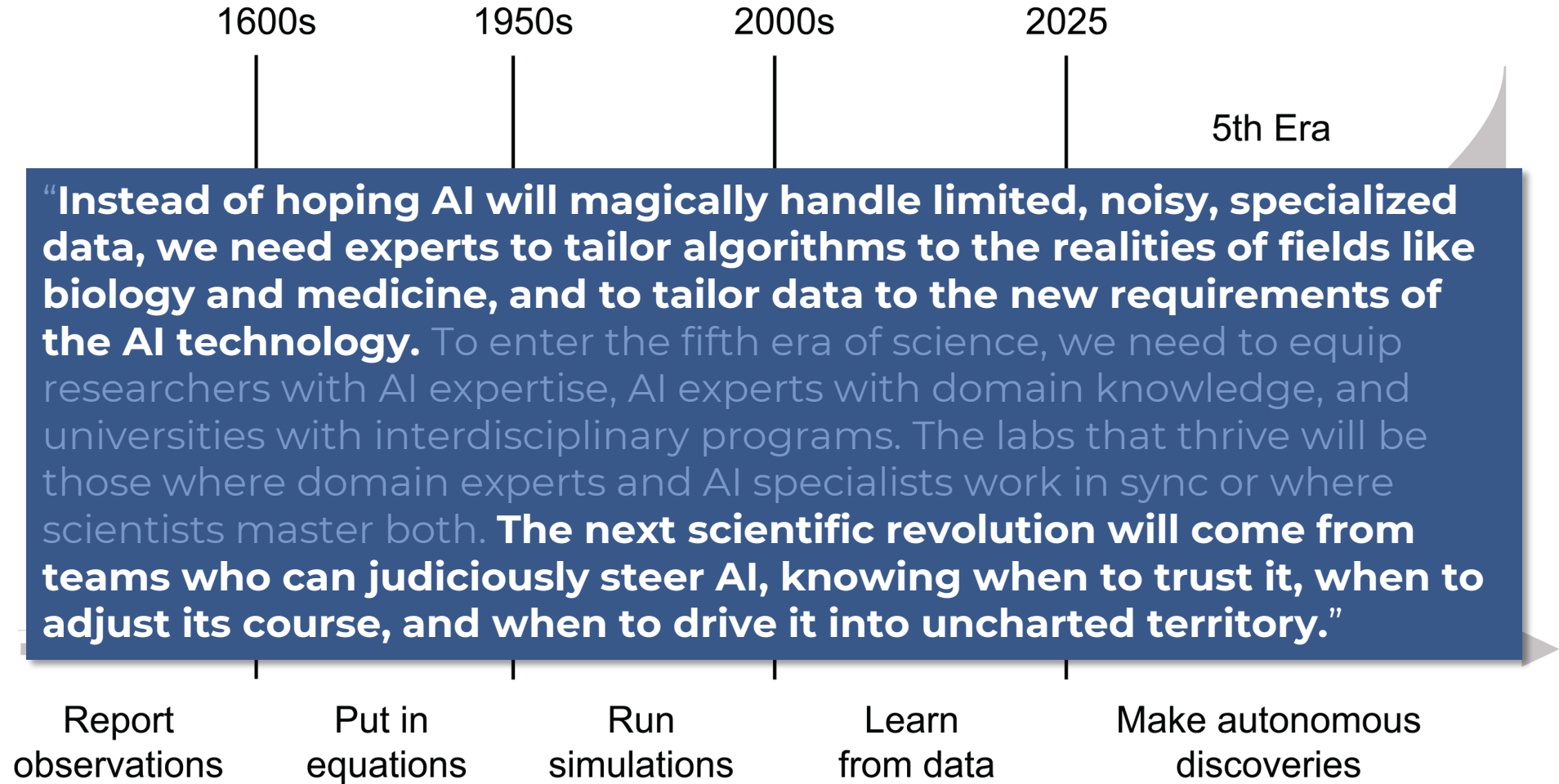
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[Miolane, 2025]



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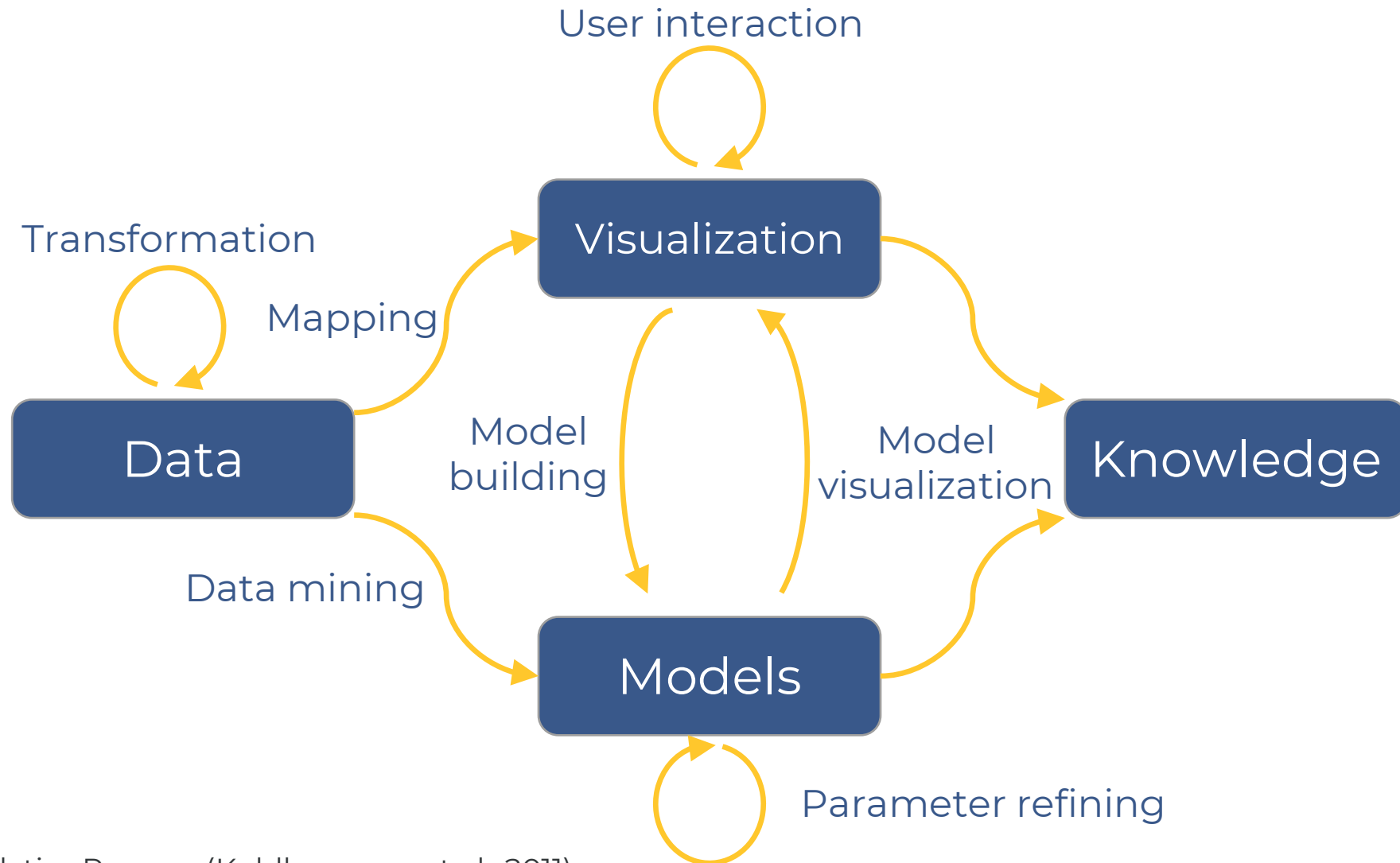


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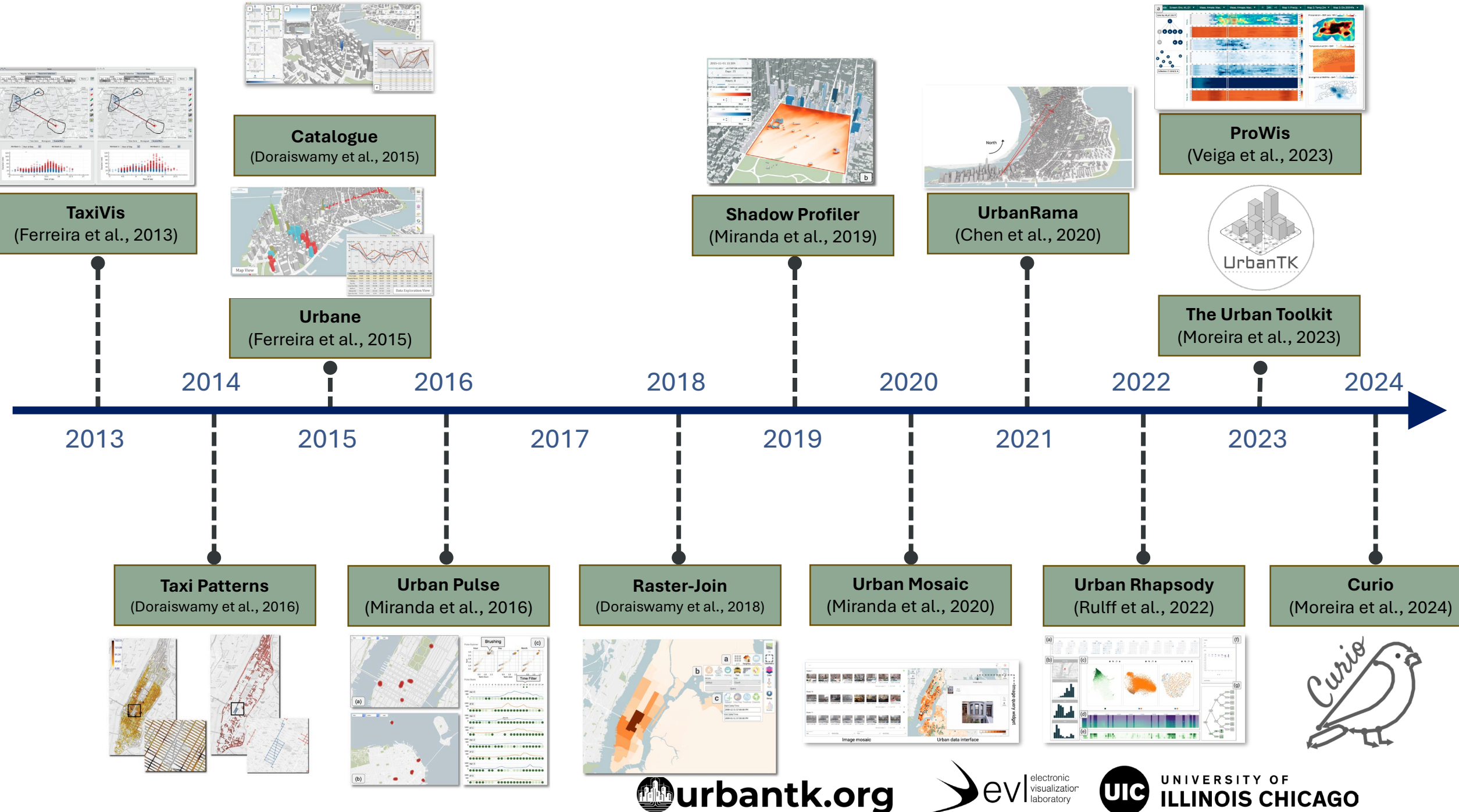
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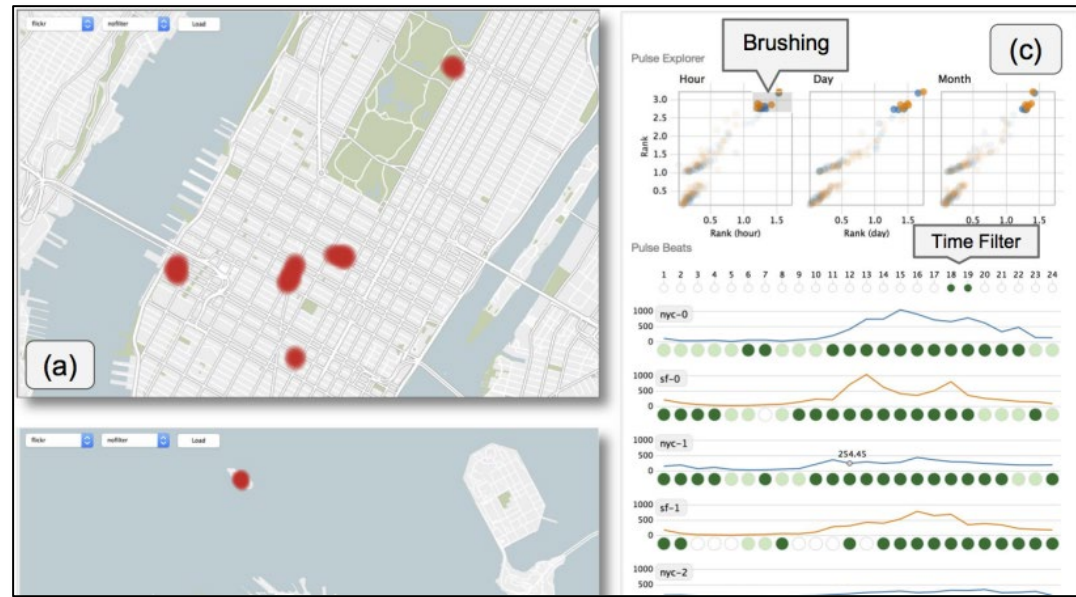
Now... what is big data visualization & analytics?



The Visual Analytics Process (Kohlhammer et al., 2011)



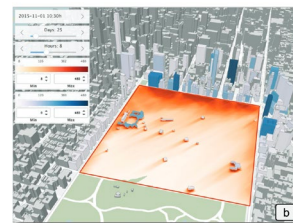




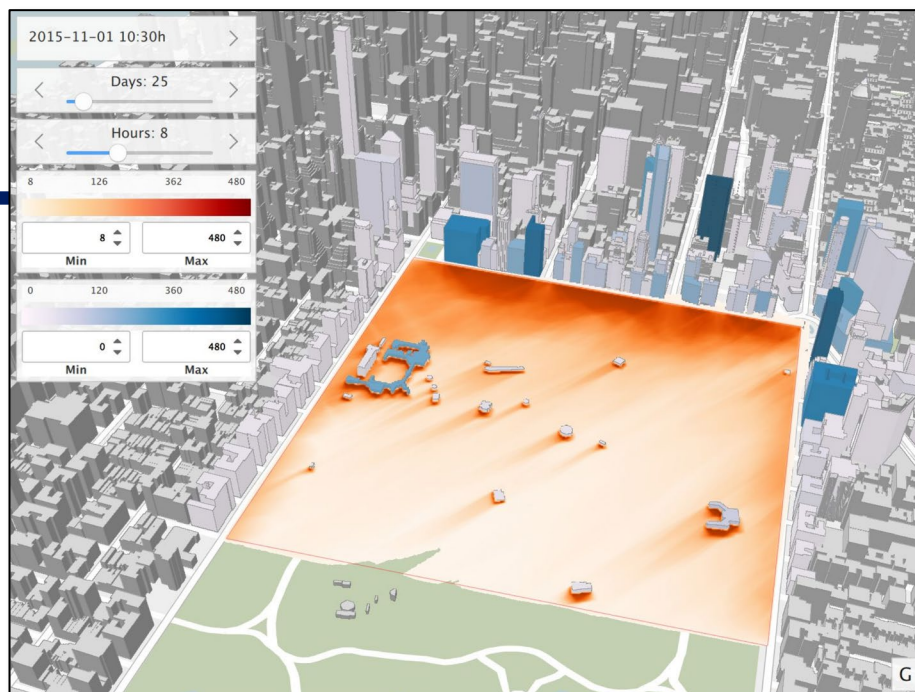
2016

Urban Pulse
(Miranda et al., 2016)

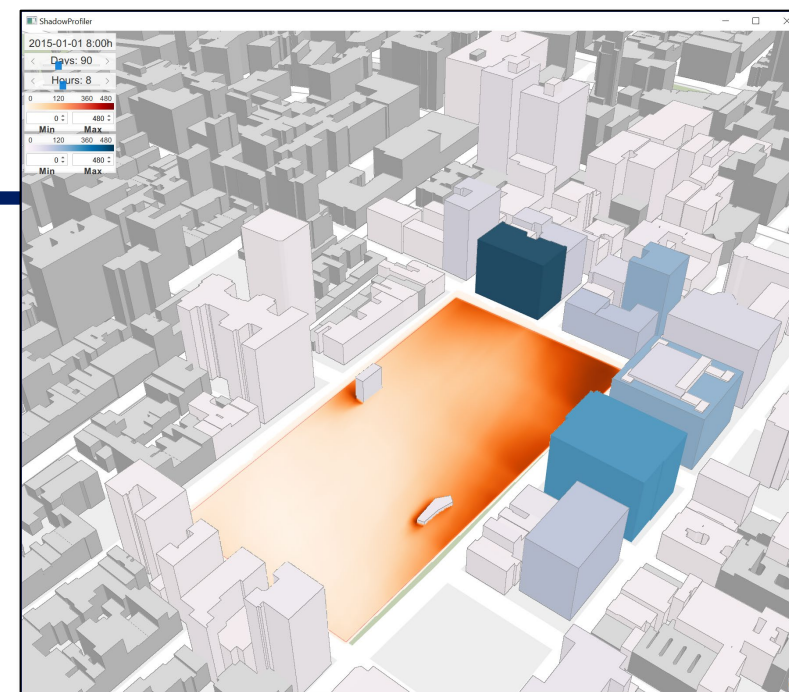


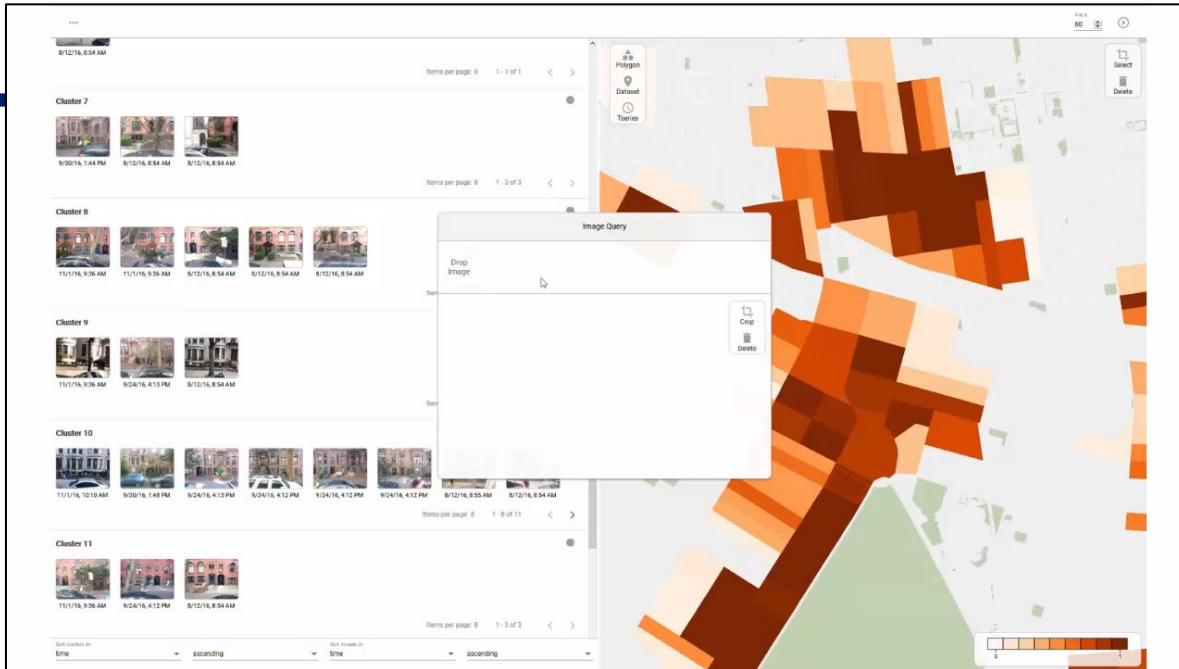
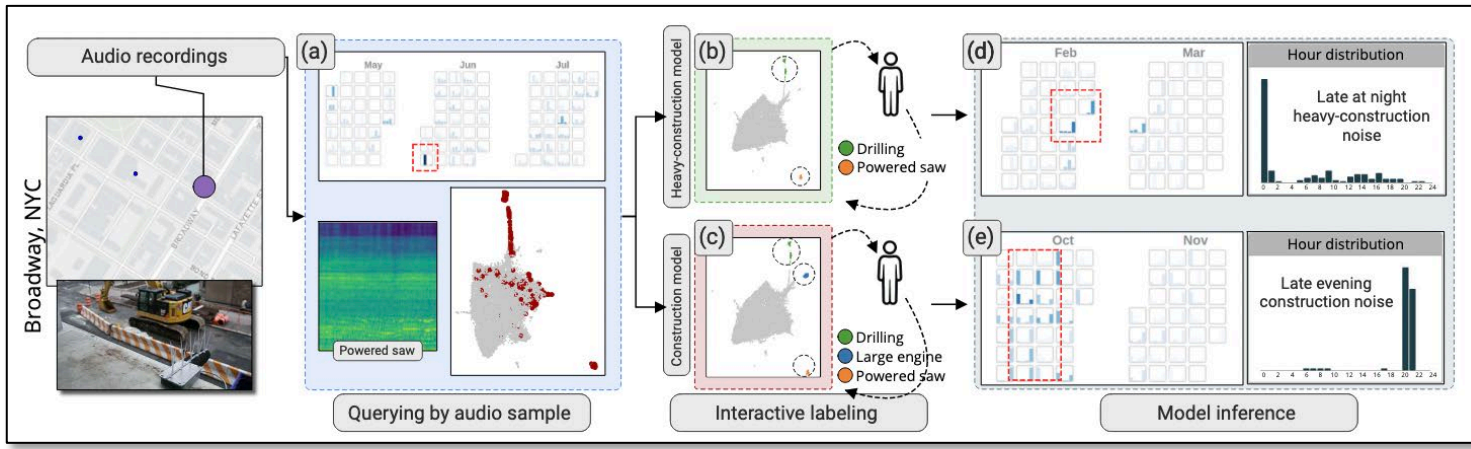


Shadow Profiler
(Miranda et al., 2019)



2019



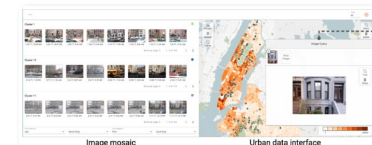


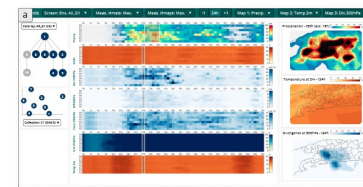
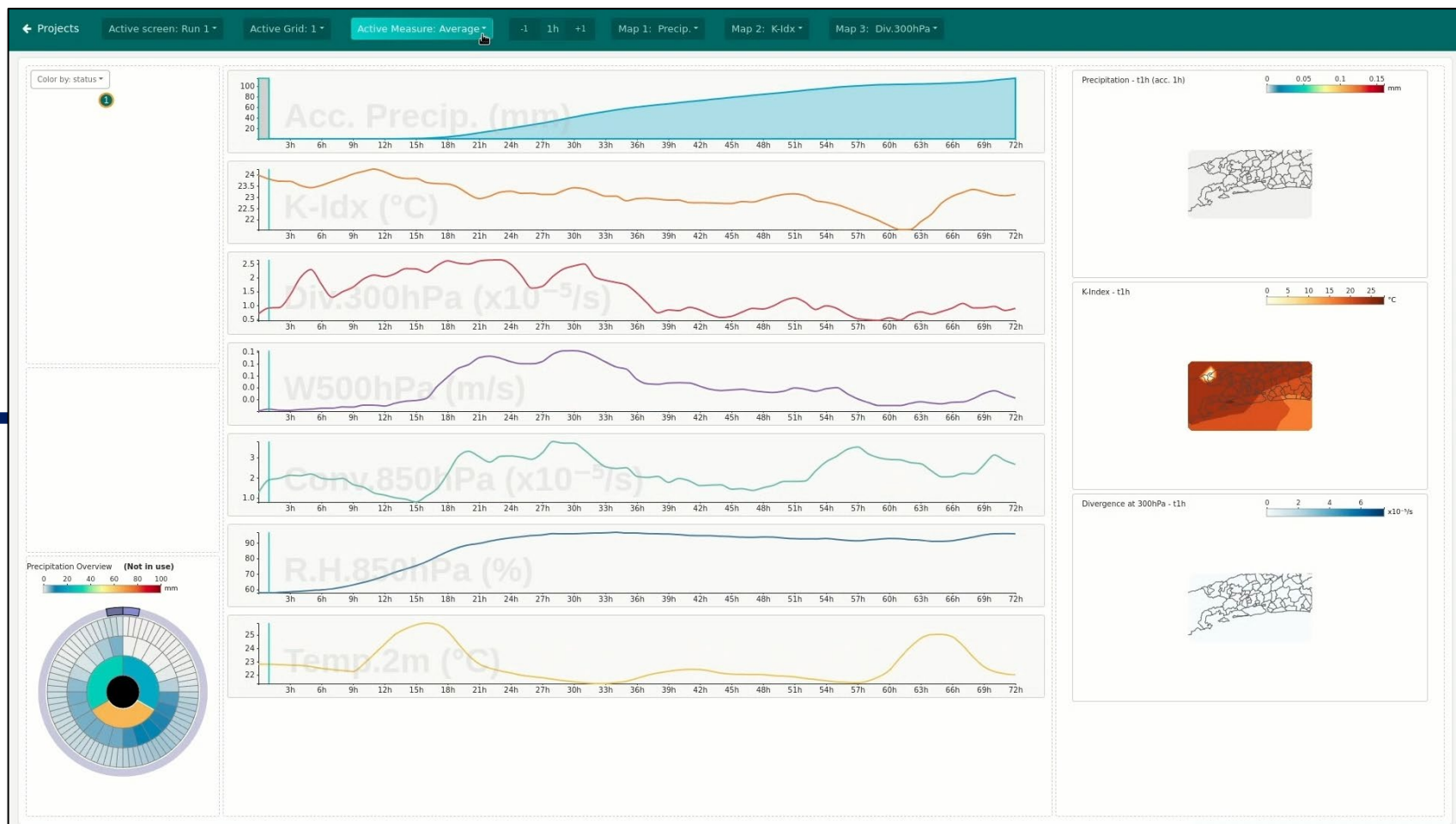
2020

2022

Urban Mosaic
(Miranda et al., 2020)

Urban Rhapsody
(Rulff et al., 2022)

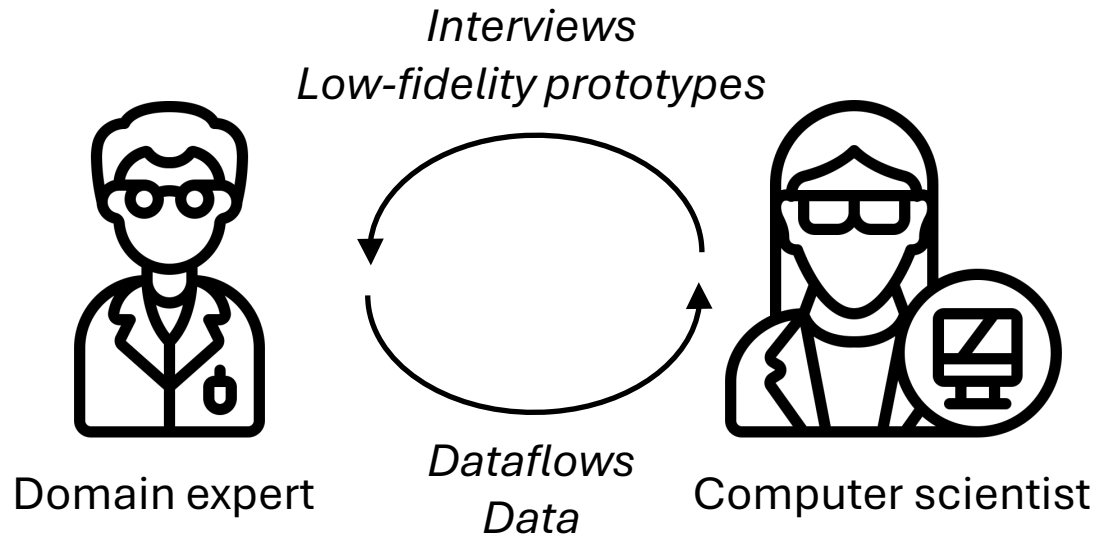




ProWis
(Veiga et al., 2023)

2023

Now... what is big data visualization & analytics?



Domain knowledge

Computer science knowledge

~30,000 lines of C++ & Java code

Urban Pulse:
Large-scale data mining of social media data

~40,000 lines of C++ code

Shadow Profiler:
City-scale assessment of sunlight access

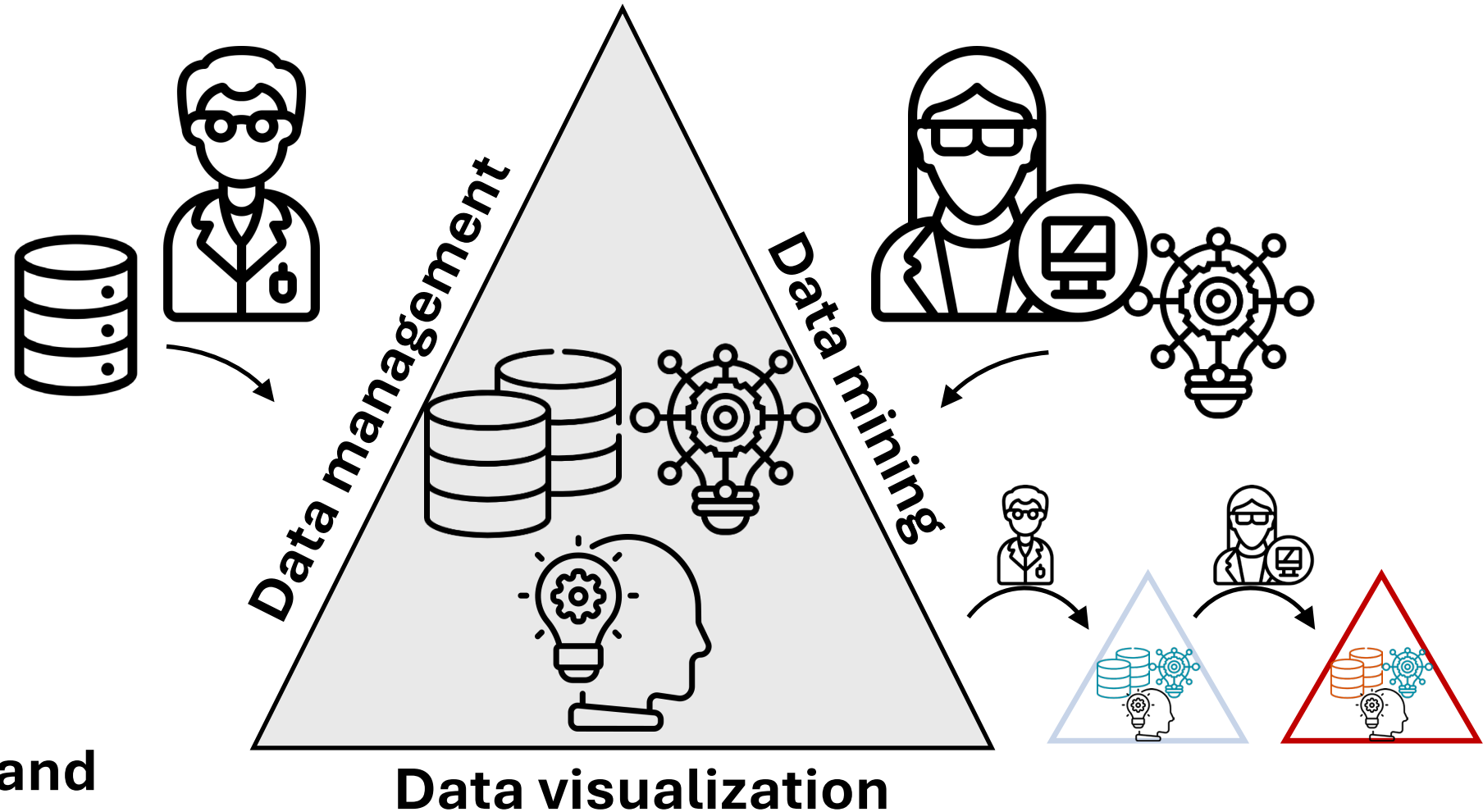
~30,000 lines of C++ code

Urban Mosaic:
Interactive exploration of large imagery data

~50,000 lines of C++ code

Urbane:
Interactive exploration of large data

Now... what is big data visualization & analytics?



Open,
transparent, and
reproducible
systems

CS 424 & CS 524: What is the difference?

424

- Visualization design
- Principles of visualization
- Task abstraction
- Visualization grammars
- Uncertainty visualization
- Data structures for visualization
- Embeddings for visualization
- ML4Vis & Vis4ML

524

- Data-to-vis systems foundations
- Building and evaluating systems in the age of AI
- Pipelines + representations for exploration



CS 524: Themes

Data-to-viz
systems
foundations

- How visualization is delivered in an end-to-end fashion when the data lives "behind" a system.

Building and
evaluating systems
in the age of AI

- Modern system-building and evaluation practices when LLMs / agents change the tooling, workflows, and architecture.

Pipelines +
representations for
exploration

- Techniques for large data summarization and to support real-time visual data exploration.

Theme 1: Data-to-viz systems foundations

- Session 1: Introduction to the course
 - Introduction to the course, goals, expectations, key themes, what is big data, EVL overview.
 - **Main goal:** Give you a high-level understanding of what it takes to build systems for big data visualization & analytics.

738

IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS, VOL. 30, NO. 1, JANUARY 2024

PROWIS: A Visual Approach for Building, Managing, and Analyzing Weather Simulation Ensembles at Runtime

Carolina Veiga Ferreira de Souza, Suzanna Maria Bonnet,
Daniel de Oliveira, Marcio Cataldi, Fabio Miranda, and Marcos Lage

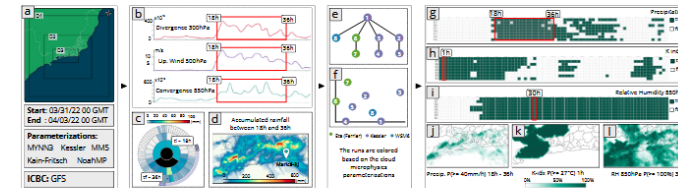


Fig. 1: Building, managing, and analyzing a Weather Research and Forecasting (WRF) model ensemble using ProWIS to study an extreme rainfall event in Maricá (RJ, Brazil) in April 2022. Before running the first simulation, weather experts defined the simulation domain, a 72-hour time horizon, the initial and boundary conditions data, and a set of physical parameterizations (a). After running the first simulation, using the line charts panel, they identified relevant predictions of divergence at 300 hPa, convergence at 850 hPa, and vertical wind at 500 hPa between 18h and 36h (b). By selecting this time interval in the sunburst chart (c), they saw that the model predicted heavy rain for several locations but underestimated the volume in the city (white marker in (d)). Using the runs overview graph (e), the experts generated a simulation ensemble for a broader study. No pattern among the members was observed in the rain volume results (f). The experts then inspected the heat matrices to verify the chances of having a minimum rainfall of 40 mm/h between 18h and 36h (g). However, in this period, the probability of this scenario was small in Maricá (i). Similarly, a minimum K-index of 27°C until 27h (h) and relative humidity of 100% (j), two conditions related to extreme rainfall, could be observed only outside the city (k, l). The results show that the model underestimated the event's disaster potential in Maricá.

Abstract—Weather forecasting is essential for decision-making and is usually performed using numerical modeling. Numerical weather models, in turn, are complex tools that require specialized training and laborious setup and are challenging even for weather experts. Moreover, weather simulations are data-intensive computations and may take hours to days to complete. When the simulation is finished, the experts face challenges analyzing its outputs, a large mass of spatiotemporal and multivariate data. From the simulation setup to the analysis of results, working with weather simulations involves several manual and error-prone steps. The complexity of the problem increases exponentially when the experts must deal with ensembles of simulations, a frequent task in their daily duties. To tackle these challenges, we propose ProWIS: an interactive and provenance-oriented system to help weather experts build, manage, and analyze simulation ensembles at runtime. Our system follows a human-in-the-loop approach to enable the exploration of multiple atmospheric variables and weather scenarios. ProWIS was built in close collaboration with weather experts, and we demonstrate its effectiveness by presenting two case studies of rainfall events in Brazil.

Index Terms—Weather visualization, Ensemble visualization, Provenance management, WRF visual setup

1 INTRODUCTION

Weather conditions significantly impact agriculture, transportation, public safety (e.g., during extreme climate events), and many other critical areas, making forecasting indispensable to designing operational strategies, enabling weather-resilient services, helping decision-making, and defining public policies [16, 34]. Although accurate weather forecasting is crucial, it is still a challenging and active research topic. To

perform weather predictions, climate specialists must consider an enormous amount of data (e.g., from satellites, weather stations), construct ensembles of numerical simulations, and rely on past experience.

Ensembles of numerical weather simulations are computed using mathematical models that describe atmospheric behavior through equations based on physical laws [31]. These models depend on initial conditions, terrain characterization, parameterization of physical processes, and discretization strategies. Weather experts must go through dependent and complex steps to run a simulation. These steps and their data dependencies resemble large-scale scientific workflows [8]. First, meteorologists must define the simulation domain considering terrain characteristics, land-use information, and spatial discretization aspects. Then, they configure the time horizon and discretization to be used during the simulation. Based on the spatiotemporal setup, the user must set the initial and boundary atmospheric conditions. Finally, previous steps are considered to specify the dynamic/thermodynamic behaviors and the micro-scale processes.

Usually, setting up these steps requires editing large text files to define several parameters. Also, to run each step, the weather expert must

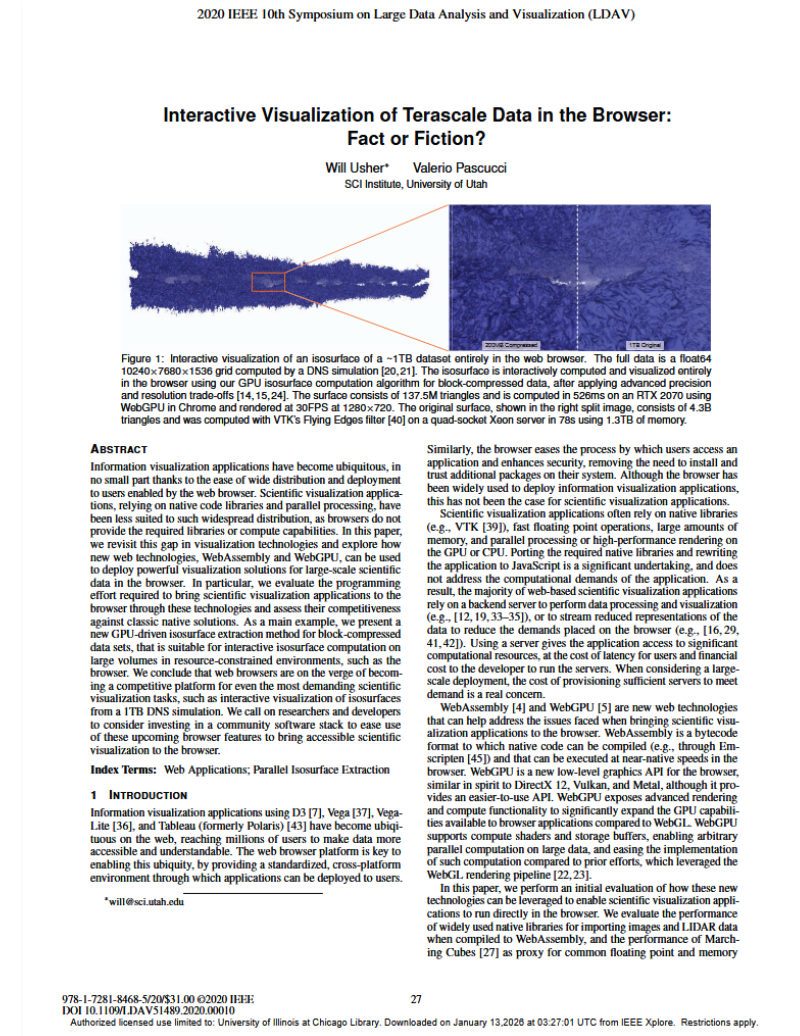
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Theme 1: Data-to-viz systems foundations

- Session 2: Web-first big-data visualization
 - How modern web stacks deliver scalable visualization when data is large and latency matters.
 - **Main goal:** Cover the latest techniques for web-based big-data visualization.



Theme 1: Data-to-viz systems foundations

- Session 3: Database-backed visualization systems
 - Database-backed visualization, where query processing, indexing, and data layout shape interactive visual performance.
 - **Main goal:** Techniques to support interactive data views of large and complex data.

436

IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS, VOL. 30, NO. 1, JANUARY 2024

Mosaic: An Architecture for Scalable & Interoperable Data Views

Jeffrey Heer and Dominik Moritz

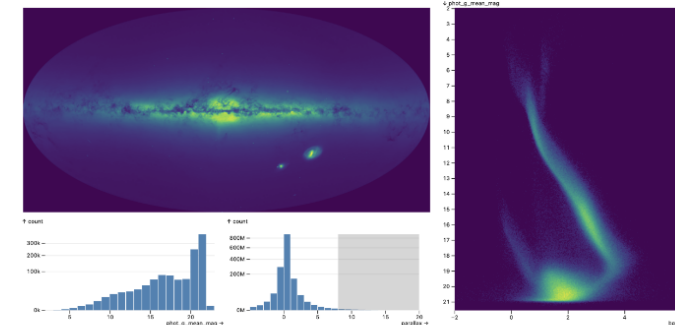


Fig. 1: A Mosaic-based interface for interactive visual exploration of all 1.8 billion stars in the Gaia star catalog. A high-resolution density map of the sky reveals our Milky Way and satellite galaxies. Stars with higher parallax values are interactively selected, forming a Hertzprung-Russell diagram of color versus stellar magnitude on the right. Mosaic offloads density and histogram computation to a backing scalable database, and automatically builds optimized data cube indexes to support interactive linked views.

Abstract—Mosaic is an architecture for greater scalability, extensibility, and interoperability of interactive data views. Mosaic decouples data processing from specification logic: clients publish their data needs as declarative queries that are then managed and automatically optimized by a coordinator that proxies access to a scalable data store. Mosaic generalizes Vega-Lite's selection abstraction to enable rich integration and linking across visualizations and components such as menus, text search, and tables. We demonstrate Mosaic's expressiveness, extensibility, and interoperability through examples that compose diverse visualization, interaction, and optimization techniques—many constructed using *vgplot*, a grammar of interactive graphics in which graphical marks act as Mosaic clients. To evaluate scalability, we present benchmark studies with order-of-magnitude performance improvements over existing web-based visualization systems—enabling flexible, real-time visual exploration of billion+ record datasets. We conclude by discussing Mosaic's potential as an open platform that bridges visualization languages, scalable visualization, and interactive data systems more broadly.

Index Terms—Visualization, Interaction, Scalability, Grammar of Graphics, Software Architecture, Databases

1 INTRODUCTION

Though many expressive visualization tools exist, scalability to large datasets and interoperability across tools remain challenging [7]. The visualization community lacks open, standardized tools for integrating visualization specifications with scalable analytic databases. While libraries like D3 [8] embrace Web standards for cross-tool interoperability, higher-level frameworks often make closed-world assumptions, complicating integration with other tools and environments.

As a concrete example, consider the Vega [36] and Vega-Lite [35] ecosystem. By default, data transformations are performed within a JavaScript runtime, limiting scalability due to both data movement and a lack of parallel computing. Meanwhile, other architectural decisions impede extensibility and interoperability. Vega-Lite's *selection*

abstraction provides a powerful, concise model for interaction, yet is realized in terms of opaque internal Vega constructs that complicate coordination with external components and environments such as notebooks. As a result, online requests¹ for table views, data-driven input widgets, more interoperable selections, and new mark types (e.g., for raster heatmaps and contour plots in Vega-Lite) have gone unaddressed for years. Other tools face similar limitations [4, 43].

We propose a standardized “middle-tier” architecture that mediates data-driven components and backing data sources. A common layer between databases and components can coordinate linked selections and parameters among views, while providing automatic query optimizations for greater scalability. We focus on the Web browser as the primary site of rendering and interaction, and seek to coordinate diverse components via standard protocols for communicating data needs, dynamic parameters, and linked selection criteria.

We contribute *Mosaic*, an architecture for interoperable, data-driven components—including visualizations, tables, and input widgets—backed by scalable data stores. A key idea of Mosaic is to decouple data processing from specification. Mosaic *Clients* communicate their data needs as declarative queries. A central *Coordinator* manages these queries, applies automatic optimizations, and pushes processing to the

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See https://www.ieee.org/publications_standards/publications/rights/index.html for more information.

¹See github.com/vega/vega/issues/ and github.com/vega/vega-lite/issues/

Theme 2: Building and evaluating systems in the age of AI

- Session 4: Benchmarking & evaluating systems
 - How to benchmark and evaluate systems rigorously, including metrics, baselines, reproducibility, and fair comparisons.
 - **Main goal:** Overview of evaluation of systems, improving reproducibility and provenance.

ReVISit 2: A Full Experiment Life Cycle User Study Framework

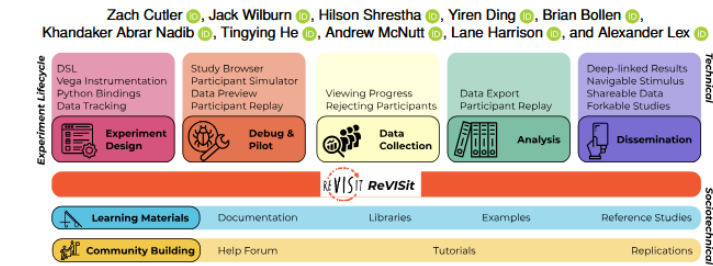


Fig. 1: ReVISit 2 supports each stage of the online user study life cycle. In addition to this linear path, the experiment life cycle is populated by internal loops between stages. For instance, issues revealed in the piloting of an experiment can lead back to the design phase to address those issues, and the process of disseminating an experiment can beget ideas for new experiments or replications.

Abstract—Online user studies of visualizations, visual encodings, and interaction techniques are ubiquitous in visualization research. Yet, designing, conducting, and analyzing studies effectively is still a major burden. Although various packages support such user studies, most solutions address only facets of the experiment life cycle, make reproducibility difficult, or do not cater to nuanced study designs or interactions. We introduce ReVISit 2, a software framework that supports visualization researchers at all stages of designing and conducting browser-based user studies. ReVISit supports researchers in the design, debug & pilot, data collection, analysis, and dissemination experiment phases by providing both technical affordances (such as replay of participant interactions) and sociotechnical aids (such as a mindfully maintained community of support). It is a proven system that can be (and has been) used in publication-quality studies—which we demonstrate through a series of experimental replications. We reflect on the design of the system via interviews and an analysis of its technical dimensions. Through this work, we seek to elevate the ease with which studies are conducted, improve the reproducibility of studies within our community, and support the construction of advanced interactive studies.

Index Terms—User studies, crowdsourcing, visualization experiments.

1 INTRODUCTION

Experimental research is a mainstay method to infer causal relationships in visualization, HCI, and related areas [19]. Experiments in visualization (often called user studies) range from perceptual studies [8, 65], to studies of visualization techniques [20, 35], studies of interaction techniques [5, 38], and studies of full visualization systems [44, 52]. Experimental approaches are also used outside of quantitative, controlled research, such as in eliciting expert feedback on systems [53] or designs [26, 37, 42], or understanding insight formation [11, 40].

While these studies historically have been conducted in the lab [8, 15, 72], they are now predominantly run online through crowd work platforms [27, 52, 63], which support rapid participant recruiting and study execution. Most desktop-focused visualization tools are designed to be accessed through the browser, suggesting that web applications are an effective means to deliver study stimuli. These trends suggest that experimental research in the future will be predominantly conducted

through browsers and asynchronously (as opposed to in a lab).

Despite the ubiquity of this form of inquiry, conducting these studies remains difficult for a variety of reasons. For instance, designing effective stimuli presents nuanced challenges (such as effective instrumentation of interactions) that are time-consuming to implement for experienced developers, let alone early-career researchers. Similarly, the design of experiments offers a maze of complex decisions (such as choice of factors, randomization, sampling strategies, and partitioning of conditions) that, when navigated poorly, can invalidate entire experiments. Compounding these design challenges is that making everything work as intended and ensuring that data is collected correctly can be tedious and error-prone.

To help with these challenges, researchers in visualization, HCI, psychology, and other fields draw on software frameworks to support their experiments. Whereas commercial tools, such as Qualtrics [29], excel at survey design, experimental design, and deployment, their closed-source and commercial nature hamper reproducibility, and they do not meet the more specialized needs of visualization experiments—such as sophisticated tracking of complex interactions. A variety of academic systems have been developed to help run studies; however, our discussions with stakeholders from the visualization community revealed that most are used only by individual research groups, use outdated technology, are difficult to maintain, and result in brittle deployments. Other academic systems are specific to certain visualization techniques (e.g., graphs [54]), or cover only a small slice of the experimental life cycle (such as factorial experiment design [50]).

- Zach Cutler, Jack Wilburn, Brian Bollen, Khandaker Abrar Nadib, Tingying He, Andrew McNutt, and Alexander Lex are with the University of Utah.
- Hilson Shrestha, Yiren Ding, and Lane Harrison are with WPI.

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Theme 2: Building and evaluating systems in the age of AI

- Session 5: VA system building
 - How system-building changes with LLMs and agents, from prototyping and debugging to architecture and reliability.
 - **Main goal:** Latest approaches to support VA system building.

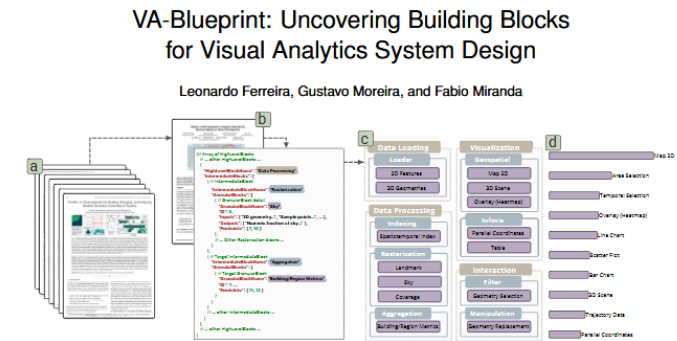


Fig. 1: VA-Blueprint provides a structured way to uncover and organize the fundamental building blocks used in visual analytics (VA) systems. (a) The process begins with the curation of a corpus of VA system research papers. (b) From each paper, core components and their relationships are systematically extracted and structured into a formal, hierarchical blueprint (represented as JSON), detailing the system's components and their dependencies. (c) This blueprint categorizes components across multiple levels of abstraction: high-level stages (e.g., **Data Processing**), intermediate groups (e.g., **Rasterization**), and specific granular blocks (e.g., **Sky Computation**). (d) The final knowledge base comprises 101 machine-readable blueprints encoding component hierarchies, and data and interaction dependencies.

Abstract—Designing and building visual analytics (VA) systems is a complex, iterative process that requires the seamless integration of data processing, analytics capabilities, and visualization techniques. While prior research has extensively examined the social and collaborative aspects of VA system authoring, the practical challenges of developing these systems remain underexplored. As a result, despite the growing number of VA systems, there are only a few structured knowledge bases to guide their design and development. To tackle this gap, we propose VA-Blueprint, a methodology and knowledge base that systematically reviews and categorizes the fundamental building blocks of urban VA systems, a domain particularly rich and representative due to its intricate data and unique problem sets. Applying this methodology to an initial set of 20 systems, we identify and organize their core components into a multi-level structure, forming an initial knowledge base with a structured blueprint for VA system development. To scale this effort, we leverage a large language model to automate the extraction of these components for other 81 papers (completing a corpus of 101 papers), assessing its effectiveness in scaling knowledge base construction. We evaluate our method through interviews with experts and a quantitative analysis of annotation metrics. Our contributions provide a deeper understanding of VA systems' composition and establish a practical foundation to support more structured, reproducible, and efficient system development. VA-Blueprint is available at urbantk.org/va-blueprint.

Index Terms—Visual analytics, large language models, knowledge base, system development, urban visual analytics.

1 INTRODUCTION

Visual analytics (VA) systems help users make sense of complex data by combining analytics with interactive visualizations. Their usefulness has been demonstrated across diverse domains, including biology [27], healthcare [28], urban planning [17], and climate science [20]. However, building VA systems remains a highly complex and iterative process. System builders must integrate data processing techniques, analytics capabilities, and visualization strategies while ensuring interpretability for domain experts and maintaining efficiency to support interactivity. Despite their significance and inherent challenges, VA

system development is often approached in an ad-hoc manner, with minimal reuse of existing components. Given the necessity to address the unique analytical and visualization needs of specific domains, these systems are usually bespoke solutions, leading to fragmented development efforts, making them difficult to extend, adapt, or reuse across projects. This tension, between the need for tailored solutions and the demand for scalable, extensible systems, presents a fundamental challenge in VA system development [12, 58]. On one hand, VA systems must be grounded by domain-specific data, tasks, and analytical workflows: a visualization technique that works well for transportation may be inadequate for urban planning, just as an analytical model for climate science may not be applicable to public health. On the other hand, as VA systems become increasingly complex, the lack of systematic and reusable components hampers the design process and increases the effort needed for development.

Currently, there is a lack of practical knowledge bases that document the building blocks of VA systems. Initial steps have produced valuable taxonomies for visualization tasks and techniques [5]. More recently, new initiatives have emerged to catalog and structure design elements

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Digital Object Identifier: xx.xxxx/TVCG.201x.xxxxxx

Theme 2: Building and evaluating systems in the age of AI

- Session 6: Intelligent agents in Visual Analytics
 - How intelligent agents can support visual analytics workflows and what effective human–AI collaboration looks like in practice.
 - **Main goal:** Methods and approaches for data exploration, discovery in the age of AI.

A Design Space for Intelligent Agents in Mixed-Initiative Visual Analytics

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RITA SEVASTJANOVA, ETH Zürich, Switzerland

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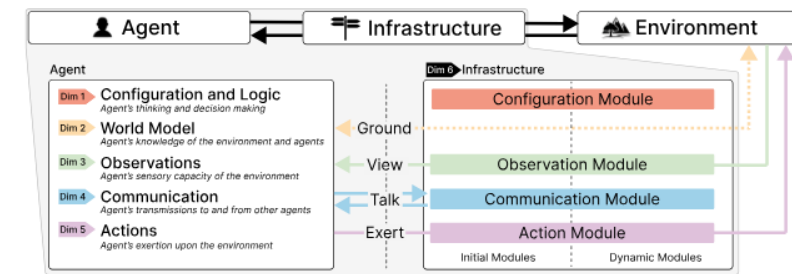


Fig. 1. To investigate the design space of intelligent agents, a mixed-initiative visual analytics system is partitioned into three components: Agent, Infrastructure, and Environment. Focusing on the agent, the design space characterizes how agents interact with the environment and other agents, given the five dimensions of the agent (Dim 1–5) – Configuration and Logic, World Model, Observations, Communication, and Actions – and the description of the Infrastructure (Dim 6).

Mixed-initiative visual analytics (VA) systems, where human and artificial intelligence (AI) agents collaborate as equal partners during analysis, represented a paradigm shift in human-computer interaction. With recent advances in AI, these systems have seen an increase in sophisticated software agents that have improved task planning, reasoning, and completion capabilities. However, while existing work characterizes agent interplay and communication strategies, there is a limited understanding of the overarching design principles for intelligent agents. Through a systematic review of 90 systems (and 207 unique agents), we propose a design space of intelligent agents comprising six dimensions that collectively characterize an agent's perception, environmental understanding, action capability, and communication strategies. We contribute a novel framework for researchers and designers to explore various design choices for new systems and to situate a system in the current landscape. We conclude with future research opportunities for intelligent agents in mixed-initiative VA systems.

CCS Concepts: • Human-centered computing → Interactive systems and tools.

Additional Key Words and Phrases: Agents, Intelligent Agent, Multi-Agents, Mixed-Initiative, Visual Analytics, Design Space

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1

Theme 3: Pipelines + representations for exploration

- Session 7: Streaming + learned + online transformation pipelines
 - Streaming and online/learned pipelines for large data.
 - **Main goal:** Techniques to support large data exploration.

NeuralCubes: Deep Representations for Visual Data Exploration

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Abstract—Visual exploration of large multi-dimensional datasets has seen tremendous progress in recent years, allowing users to express rich data queries that produce informative visual summaries, all in real time. Techniques based on data cubes are some of the most promising approaches. However, these techniques usually require a large memory footprint for large datasets. To tackle this problem, we present NeuralCubes: neural networks that predict results for aggregate queries, similar to data cubes. NeuralCubes learns a function that takes as input a given query, for instance, a geographic region and temporal interval, and outputs the result of the query. The learned function serves as a real-time, low-memory approximator for aggregation queries. Our models are small enough to be sent to the client side (e.g. the web browser for a web-based application) for evaluation, enabling data exploration of large datasets without database/network connection. We demonstrate the effectiveness of NeuralCubes through extensive experiments on a variety of datasets and discuss how NeuralCubes opens up opportunities for new types of visualization and interaction.

I. INTRODUCTION

Interactive visual exploration is becoming increasingly essential for making sense of large multi-dimensional datasets. It is common for datasets to have billions of data items that contain a variety of attributes of geographic, temporal, and categorical nature. Due to this size and complexity, issuing data queries in real-time is often not feasible as it will result in an unreasonable amount of latency. Instead, efficient data structures are often designed for real-time exploration, in lieu of making raw database queries. These data structures are pre-computed and optimized around queries frequently used by the visualization, such as performing data summaries [33], [45], ranking [38], and applying multivariate statistics [57].

However, while these data structures are effective, they can still be prohibitively large as data size increases. Worse, when data complexity increases (i.e. in terms of the number of dimensions in the data), the sizes of many of these data structures grow exponentially. As a result, these data structures



Fig. 1. NeuralCubes frames approximate query processing over a database as a neural network, one that takes a query as input, e.g. selection of a temporal interval, and maps it to an aggregation result as output, e.g. a count of the number of records, enabling real-time visual exploration for large-scale data. In the figures above, the blue lines represent visualizations generated with data from the database and the orange lines are generated using NeuralCubes. Note that the two are nearly indistinguishable, but NeuralCubes uses only a fraction of the memory footprint as the database.

are often stored on a server, and only sub-parts of the data structures are fetched in real-time based on the user's exploration.

In this paper, we introduce NeuralCubes, a technique that approximates queries on multi-dimensional datasets for the support of interactive visual exploration. NeuralCubes is a deep neural network that is compact in storage, can respond to data queries in real time, and is sufficiently accurate for visualization purposes. Thus, unlike existing systems that have high storage requirements and require intricate data structures [33], [34],

Theme 3: Pipelines + representations for exploration

- Session 8: Progressive + uncertainty visualization
 - Progressive and uncertainty visualization techniques for large data
 - **Main goal:** Overview of progressive and uncertainty visualization, as well as techniques to support it.

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6447

A Survey on Progressive Visualization

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(Survey Paper)

Abstract—Currently, growing data sources and long-running algorithms impede user attention and interaction with visual analytics applications. Progressive visualization (PV) and visual analytics (PVA) alleviate this problem by allowing immediate feedback and interaction with large datasets and complex computations, avoiding waiting for complete results by using partial results improving with time. Yet, creating a progressive visualization requires more effort than a regular visualization but also opens up new possibilities, such as steering the computations towards more relevant parts of the data, thus saving computational resources. However, there is currently no comprehensive overview of the design space for progressive visualization systems. We surveyed the related work of PV and derived a new taxonomy for progressive visualizations by systematically categorizing all PV publications that included visualizations with progressive features. Progressive visualizations can be categorized by well-known visualization taxonomies, but we also found that progressive visualizations can be distinguished by the way they manage their data processing, data domain, and visual update. Furthermore, we identified key properties such as uncertainty, steering, visual stability, and real-time processing that are significantly different with progressive applications. We also collected evaluation methodologies reported by the publications and conclude with statistical findings, research gaps, and open challenges.

Index Terms—Progressive visual analytics, progressive visualization, taxonomy, state-of-the-art report, survey.

1. INTRODUCTION

THE research field of progressive visualization (PV) and visual analytics (PVA) have gained increasing attention in the past years to use more complex algorithms processed over larger data sources. Although the name progressive visual

analytics is relatively new, the core concept of creating a progressively improving representation of large data or time-intensive algorithms was already introduced over 20 years ago. PVA can be split into data processing and visual-interactive parts. While data processing has been the main subject of several studies, e.g., [21], [34], visualization aspects, on the contrary, received less attention. This survey focuses on the design of progressive visualizations. Earlier research for this concept is also known as incremental visualization [23], online visualization [34], fine-grain visualization [88], progressive visualization [25], iteration visualization [49], optimistic visualization [64], or approximate visualization [52]. This shows that the research field is applied in many domains but is not completely defined and still has many open questions that have to be explored. One of these questions is what makes a visualization progressive compared to traditional, non-progressive (*instantaneous, monolithic, or eager* [5]) visualizations. Therefore, we decided to gather all PVA-related publications which provide a progressive visualization for data analytics and created the first survey in this research area.

As we will see in this survey, starting in the 2000s, large multidimensional datasets required new approaches, and the first progressive visualization systems were created. Although early solutions were limited to single visualization categories, many of the ideas in terms of data processing and progressive key properties were already sophisticated. In the 2010s, the publication rate raised significantly, and more compound PVA systems were introduced, supporting domain experts with specific data analytics tasks. A greater variety of large datasets emerged in several domains, including urban science, healthcare, IT and communication, and meteorology. The demand for tailored solutions resulted in more expert evaluations and case studies to measure the impact of contributions. This trend held on during the start of the 2020s as there are still many open research questions for PVA. However, there is currently no comprehensive summary structuring and discussing the different progressive visualization approaches.

With this survey, we intend to fill this gap. The design of progressive algorithms covers data chunking, reuse of previous partial results, and quality estimation. However, the design of progressive visualizations tackles additional complications, as for example, the visual stability between updates, the role of user interaction steering the process, and the conveyance of uncertainty. Hence, we primarily focus on those challenges that result from a user being tightly involved in the process via

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A continuously updated visual browser of the survey data is available at visualsurvey.net/pva.

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Theme 3: Pipelines + representations for exploration

- Session 9: Computational topology
 - Computational topology techniques for summarizing and reasoning about shape/structure in complex data.
 - **Main goal:** Overview of topology techniques and toolkits.

832

IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS, VOL. 24, NO. 1, JANUARY 2018

The Topology ToolKit

Julien Tierny, Guillaume Favelier, Joshua A. Levine, *Member, IEEE*, Charles Gueunet, and Michael Michaux

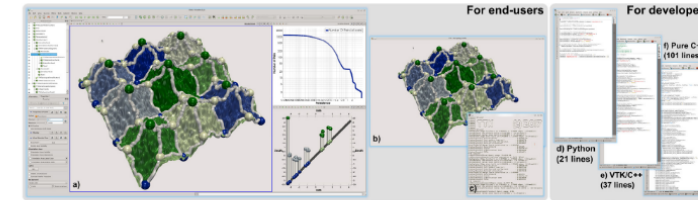


Fig. 1. TTK is a software platform for the topological analysis of scalar data in scientific visualization. It is both easily accessible to end users (with ParaView plugins (a), VTK-based generic GUIs (b) or command-line programs (c)) and flexible for developers (Python (d), VTK/C++ (e) or dependency-free C++ (f) bindings). TTK provides an efficient and unified approach to topological data representation and simplification, which enables in this example a discrete Morse-Smale complex (a) to comply to the level of simplification dictated by a piecewise linear persistence diagram (bottom-right linked view, a). Code snippets are provided (d-f) to reproduce this pipeline.

Abstract—This system paper presents the Topology ToolKit (TTK), a software platform designed for the topological analysis of scalar data in scientific visualization. While topological data analysis has gained in popularity over the last two decades, it has not yet been widely adopted as a standard data analysis tool for end users or developers. TTK aims at addressing this problem by providing a unified, generic, efficient, and robust implementation of key algorithms for the topological analysis of scalar data, including: critical points, integral lines, persistence diagrams, persistence curves, merge trees, contour trees, Morse-Smale complexes, fiber surfaces, continuous scatterplots, Jacobi sets, Reeb spaces, and more. TTK is easily accessible to end users due to a tight integration with ParaView. It is also easily accessible to developers through a variety of bindings (Python, VTK/C++) for fast prototyping or through direct, dependency-free, C++, to ease integration into pre-existing complex systems. While developing TTK, we faced several algorithmic and software engineering challenges, which we document in this paper. In particular, we present an algorithm for the construction of a discrete gradient that complies to the critical points extracted in the piecewise-linear setting. This algorithm guarantees a combinatorial consistency across the topological abstractions supported by TTK, and importantly, a unified implementation of topological data simplification for multi-scale exploration and analysis. We also present a cached triangulation data structure, that supports time efficient and generic traversals, which self-adjusts its memory usage on demand for input simplicial meshes and which implicitly emulates a triangulation for regular grids with no memory overhead. Finally, we describe an original software architecture, which guarantees memory efficient and direct accesses to TTK features, while still allowing for researchers powerful and easy bindings and extensions. TTK is open source (BSD license) and its code, online documentation and video tutorials are available on TTK's website [108].

Index Terms—Topological data analysis, scalar data, data segmentation, feature extraction, bivariate data, uncertain data

1 INTRODUCTION

As scientific datasets become more intricate and larger in size, advanced data analysis algorithms are needed for their efficient visualization and exploration. For scalar field visualization, topological analysis techniques [69] have shown to be practical solutions in various contexts by enabling the concise and complete capture of the structure of the input data into high-level *topological abstractions* such as contour trees [29], Reeb graphs [18, 85, 109], or Morse-Smale complexes [36, 64]. Such topological abstractions are fundamental data structures that enable

the development of advanced data analysis, exploration and visualization techniques, including for instance: small seed set extraction for fast isosurface traversal [30, 111], feature tracking [101], transfer function design for volume rendering [113], similarity estimation [105], or application-driven segmentation and analysis tasks [63, 67, 68, 75]. Successful applications in a variety of fields of science, including combustion [24, 63, 75], material sciences [67, 68], chemistry [57], or astrophysics [97, 102] to name a few, have even been documented, which further stresses the importance of this class of techniques. Despite this popularity and success in applications, topological data analysis (TDA) has not yet been widely adopted as a standard data analysis tool for end users and developers. While some open source implementations for specific algorithms are available [31, 39, 42, 43, 56, 96, 102, 106], we still identify three main issues preventing a wider adoption of TDA.

First, these implementations lack, in general, support for standard data file formats, genericity regarding the dimensionality of the input data, integration into graphical user front ends, or access through high-level scripting languages. These limitations challenge their adoption by end users and domain experts with little or no programming knowledge.

Second, regarding software developers, each implementation comes with its own internal data structures or its own list of third-party software dependencies, which challenges their integration into pre-existing, complex systems for visualization or data analysis.

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Theme 3: Pipelines + representations for exploration

- Session 10: Computational frameworks for large-scale visual analytics
 - Neural fields, encoder-decoder models, and other advanced techniques to support large-scale visual analytics.
- **Main goal:** Latest techniques to support large-scale visual analytics.

A Neural Field-Based Approach for View Computation & Data Exploration in 3D Urban Environments

Stefan Cobeli , Kazi Shahrukh Omar , Rodrigo Valença, Nivan Ferreira , and Fabio Miranda 

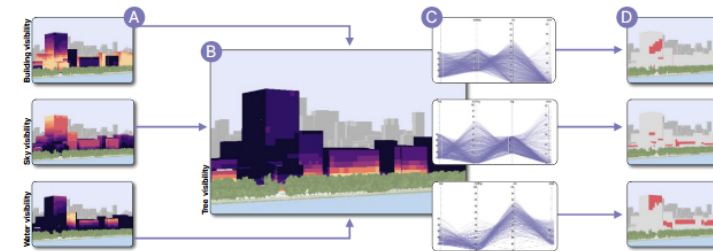


Fig. 1: Direct queries for the computation of thematic data associated with building facades representing the visibility towards buildings, sky, and water, and the visibility towards trees. Following the user specified conditions in the parallel coordinates we perform inverse queries to infer the building facade positions satisfying the joint visibility constraints.

Abstract—Despite the growing availability of 3D urban datasets, extracting insights remains challenging due to computational bottlenecks and the complexity of interacting with data. In fact, the intricate geometry of 3D urban environments results in high degrees of occlusion and requires extensive manual viewpoint adjustments that make large-scale exploration inefficient. To address this, we propose a view-based approach for 3D data exploration, where a vector field encodes views from the environment. To support this approach, we introduce a neural field-based method that constructs an efficient implicit representation of 3D environments. This representation enables both faster *direct queries*, which consist of the computation of view assessment indices, and *inverse queries*, which help avoid occlusion and facilitate the search for views that match desired data patterns. Our approach supports key urban analysis tasks such as visibility assessments, solar exposure evaluation, and assessing the visual impact of new developments. We validate our method through quantitative experiments, case studies informed by real-world urban challenges, and feedback from domain experts. Results show its effectiveness in finding desirable viewpoints, analyzing building facade visibility, and evaluating views from outdoor spaces. Code and data are publicly available at urbantk.org/neural-3d.

Index Terms—Urban computing, urban visual analytics, view computation, neural field

1 INTRODUCTION

The increasing availability of urban datasets with 3D spatial components has created new opportunities for analyzing complex urban phenomena [28]. Among these possibilities are applications such as visibility evaluation [50], view analysis of environmental features [46, 52, 54], visibility planning [12, 36], view preferences [18, 38], and building design [10, 43], which often rely on computing view indices or identifying viewpoints that satisfy specific criteria. Conventional methods for computing these indices typically rely on ray casting [27] or rasterization [12], both of which are computationally expensive and scale poorly to large urban settings. To mitigate this, many studies either perform offline analyses [12, 28, 40] or rely on pre-computation to enable interactive exploration [10]; however, both approaches ultimately limit true interactivity in visual analysis. Beyond computation, exploring and interacting with 3D urban datasets presents additional difficulties. Spatial occlusion and complex camera controls often hinder interpretation:

once indices are obtained, they are commonly visualized as thematic overlays on physical surfaces (Fig. 1(A,B)). Unlike 2D visualizations with fixed perspectives, 3D exploration requires careful viewpoint selection and manipulation across multiple degrees of freedom. Without appropriate support, users may struggle to efficiently navigate, reveal patterns, and interpret the data. Although prior research has extensively explored methods for navigating and visualizing 3D urban environments, much of this work has focused on the representations of cities rather than the interactive analysis of 3D urban data itself [19]. For instance, recent studies have investigated viewpoint selection [3], occlusion-aware rendering [11], and deformation-based de-occlusion and navigation support [5]. However, these approaches only consider physical data, such as building geometry and infrastructure, while overlooking the integration of thematic attributes. As a result, two intertwined challenges arise: (1) the **efficient computation** of view-related measures that capture the relationships between physical elements (e.g., building geometry, terrain) and thematic attributes (e.g., sunlight access, visibility), and (2) the **effective exploration** of such measures in 3D environments, where interaction and interpretation are inherently more complex than in 2D settings.

To address these challenges, we propose a view-based approach for the interactive exploration of physical and thematic dimensions in this work. At the core of our method is the modeling of views as a dense multi-dimensional field, where each viewpoint is associated with a semantic attribute vector that encodes relevant properties across large

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Course schedule

- Course themes are spread across 13 weeks through 10 sessions.
 - Including 8 class meetings for research presentations.
 - **Important: Sessions are not aligned one-to-one with weeks.**
- Two weeks for the project proposal and project presentations.
- Two key weeks:
 - Week 5: Proposal presentations
 - Week 15: Project presentations

Evaluation and grading

- Research paper presentation: 30% (15% each)
- Project proposal: 10%
- Project midterm review: 15%
- Final project deliverables: 30%
- Participation: 15%

Evaluation and grading: Research presentations

- For each research presentation, you will first need to select **at least** one research paper from a top journal or conference.
- Then, you should prepare and deliver a presentation that (1) gives an overview of the papers' topic and (2) covers the papers' contributions.
- You should prepare a presentation (and lead a discussion) covering the topic and the selected papers. Your team is responsible for creating an outline that does justice to the papers.

Evaluation and grading: Project

- You will need to demonstrate your research skills by identifying, defining, and tackling a research problem within a course-related topic.
- Two tracks to success:
 1. **A pure research project:** Scope a research problem and contribute towards a scholarly solution.
 2. **A practical project:** Build a functional, “high-impact” tool or methodology that provides immediate utility.



Evaluation and grading: Project

- You will need to demonstrate your research skills by identifying, defining, and tackling a research problem within a course-related topic.
- Three milestones:
 - Milestone 1: Project proposal
 - Milestone 1 is where your team commits to a project direction that is feasible, data-backed, and technically meaningful. Think of this as a lightweight "engineering + research pitch": you should convince us you can build something non-trivial, evaluate it, and explain why it matters.
 - Milestone 2: Midterm review
 - In the second milestone, you will present your proposal and intermediate results. The goal is to show that your project is no longer just an idea: you have a working pipeline and early results that reduce risk for the rest of the semester.
 - Milestone 3: Final report and presentation
 - Milestone 3 is your final delivery: a clear story, a working artifact, and evidence that your approach solves a real problem better than obvious alternatives. Treat this like a mini conference-style submission: demo + paper + reproducible project page.



CS 424 & CS 524: Different paths to success

424

Data: Small dataset.

Outcome: Dashboard-level interface.

One path to success: Build a reasonable interface, highlighting data findings.

CS 424 & CS 524: Different paths to success

424

Data: Small dataset.

Outcome: Dashboard-level interface.

One path to success: Build a reasonable interface, highlighting data findings.

524

Data: Large or complex data.

Outcome: Technique, methodology.

Two paths to success: (1) Scope and tackle an interesting research problem; or (2) Build something practical and useful.



CS 524: Logistics

- We meet Tue and Thu 8 AM.
- Office hours: Friday 10-12 PM by appointment.
 - Send me an email by Thursday 5 PM, I will send you a calendar invite for a slot on Friday between 10 AM and 12 PM.
 - Attend office hours!

Research paper presentation: 30% (15% each)

Project proposal: 10%

Project midterm review: 15%

Final project deliverables: 30%

Participation: 15%

